

Imaging System

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A solid orange horizontal bar spanning the entire width of the slide at the bottom.

Mathematical Model of Imaging System

➤ If an object is placed in front of the film, can an image of that object be obtained?

object

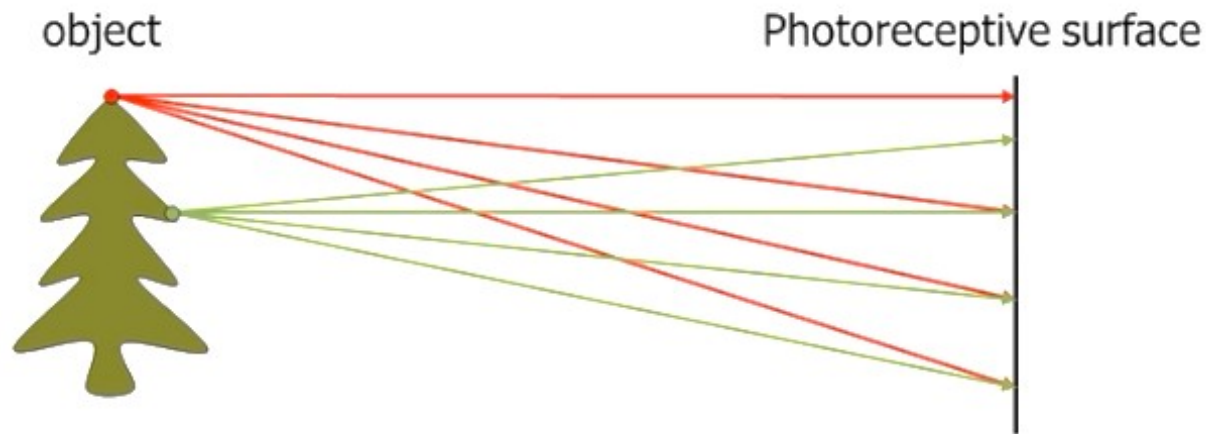


Photoreceptive surface



Mathematical Model of Imaging System

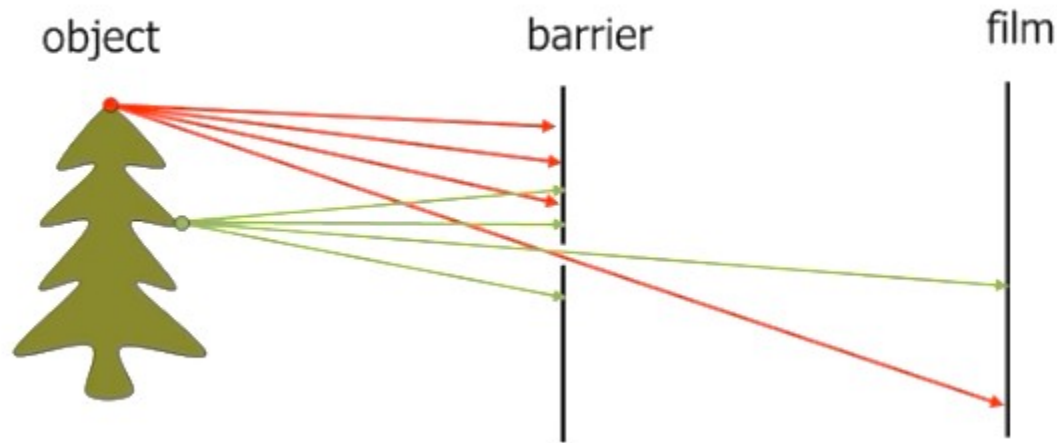
➤ If an object is placed in front of the film, can an image of that object be obtained?



Mathematical Model of Imaging System

➤ If an object is placed in front of the film, can its image be obtained?

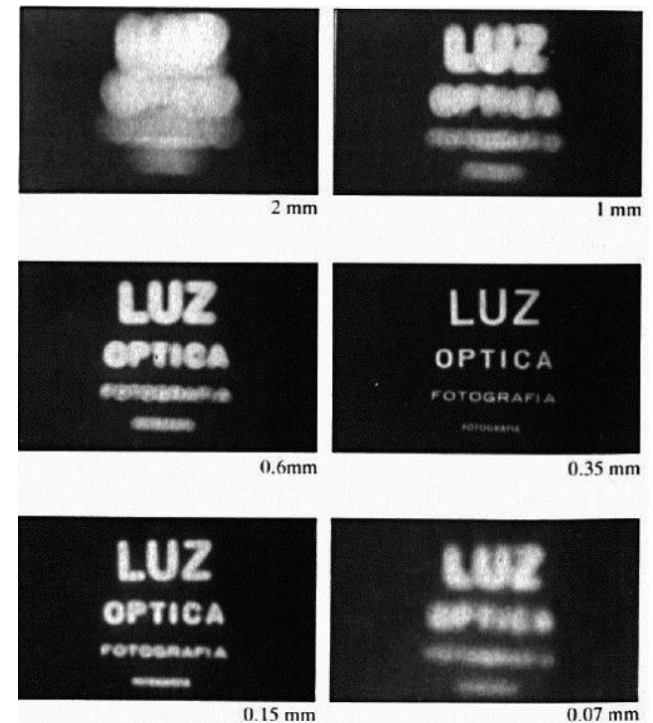
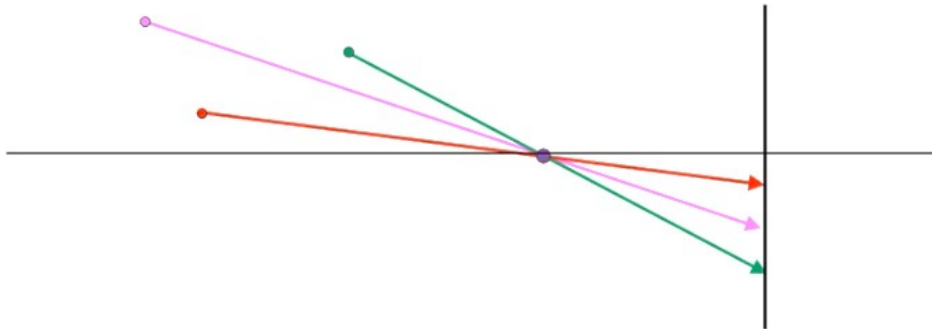
- Adding an obstacle with a pinhole can reduce the blur on the image plane.
- This pinhole is called the aperture.



Mathematical Model of Imaging System

➤ Ideal pinhole: Only one ray of light reaches the film

- The image can be very dim
- Produces diffraction effects

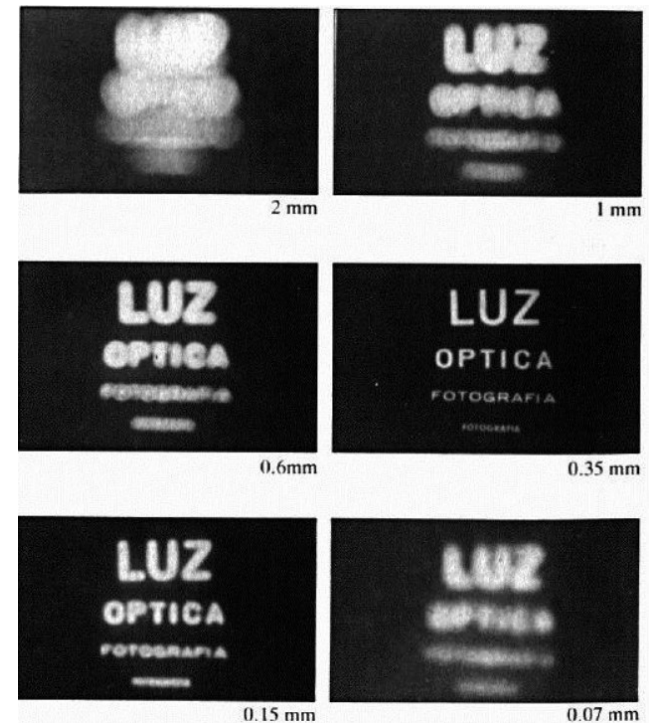
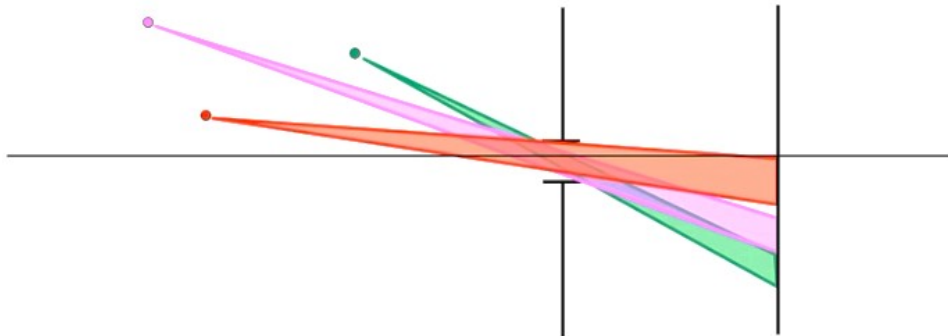


Mathematical Model of Imaging System

➤ Ideal pinhole: Only one ray of light reaches the film

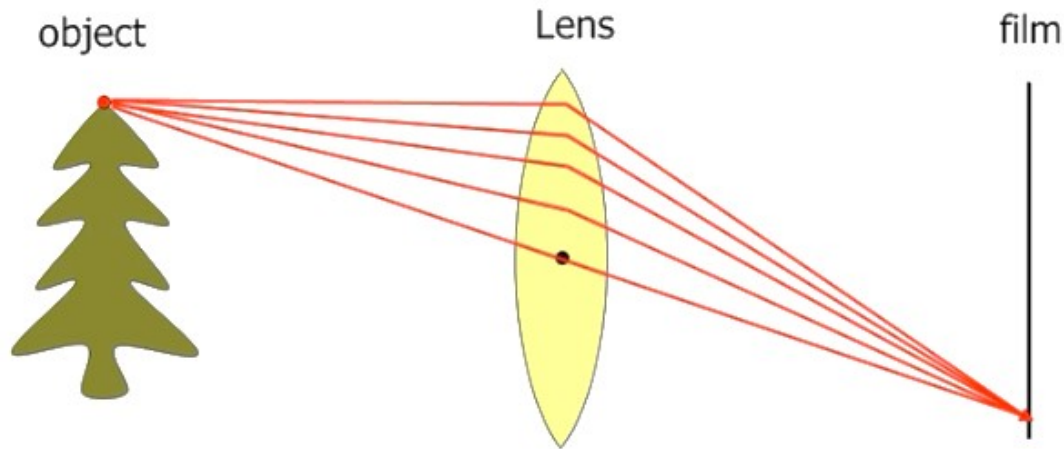
- The image can be very dim
- Produces diffraction effects

➤ Making the pinhole larger (i.e., increasing the aperture) will blur the image.



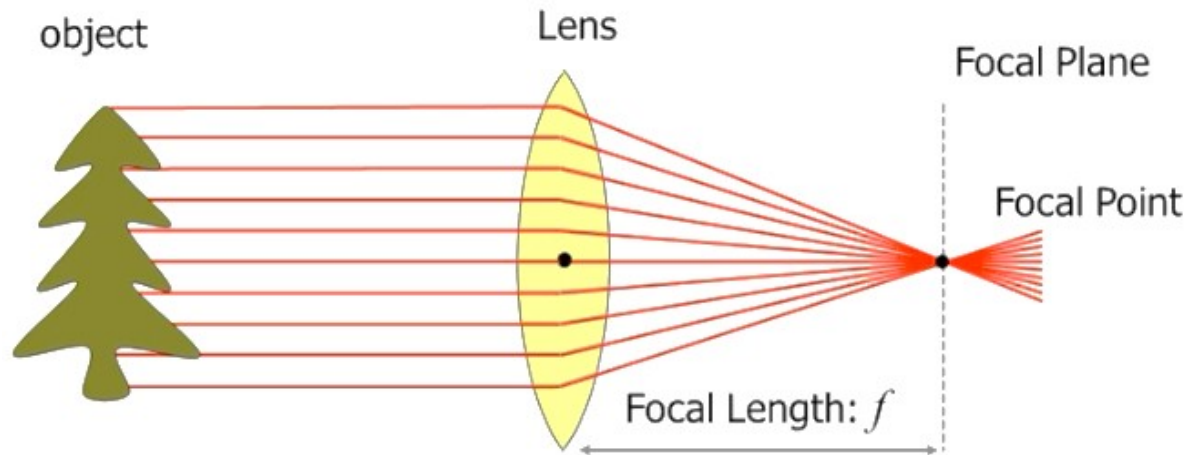
Mathematical Model of Imaging System

- The lens focuses light onto the image plane.

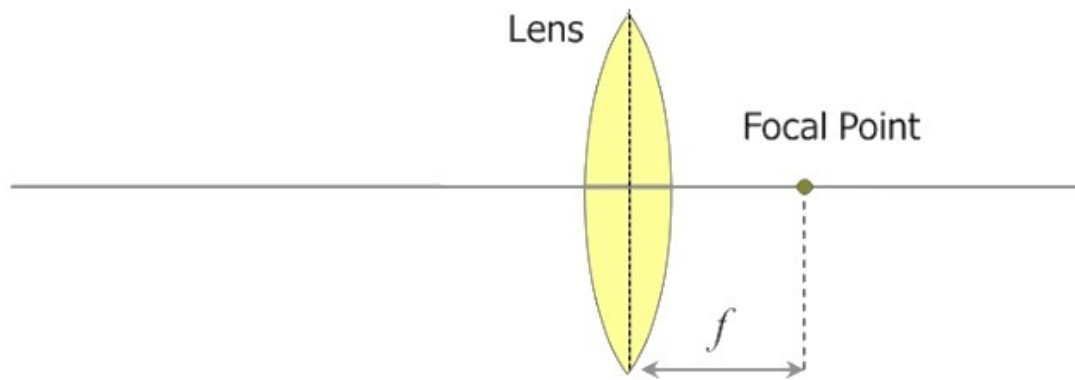


Mathematical Model of Imaging System

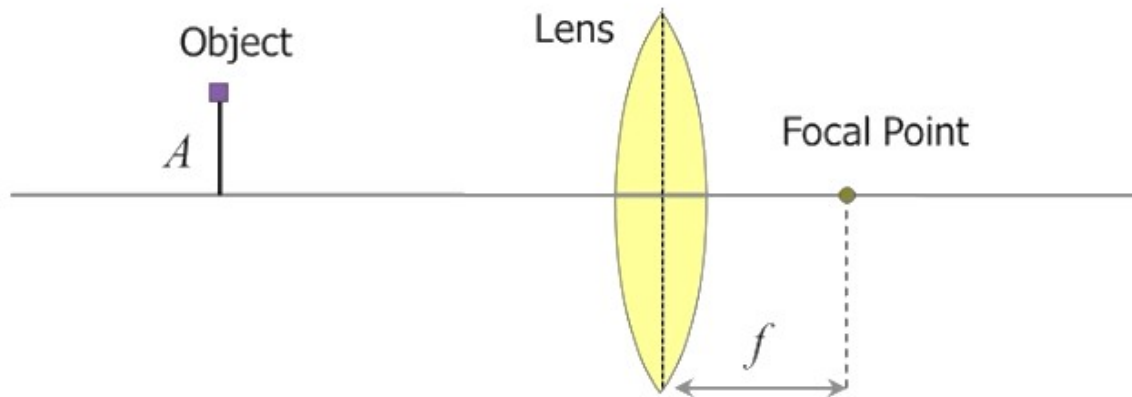
- The lens focuses light onto the image plane.
- Light rays passing through the optical center travel in a straight line.
- All light rays parallel to the optical axis converge at the focal point.



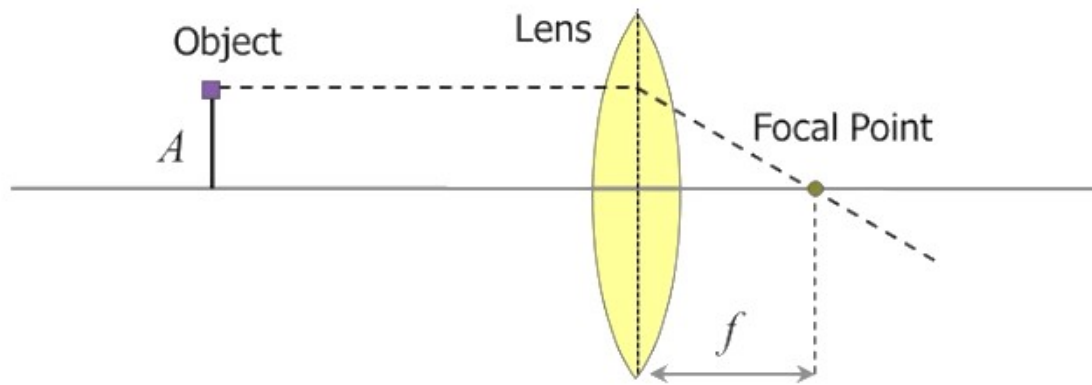
Mathematical Model of Imaging System



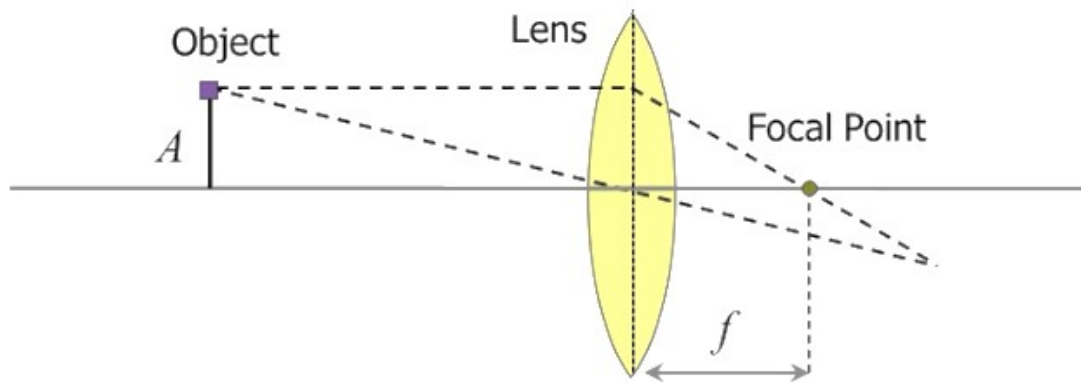
Mathematical Model of Imaging System



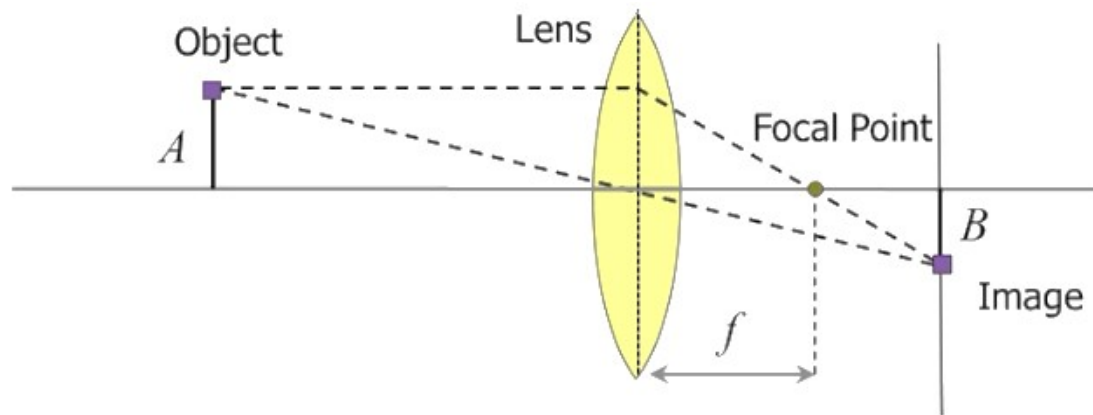
Mathematical Model of Imaging System



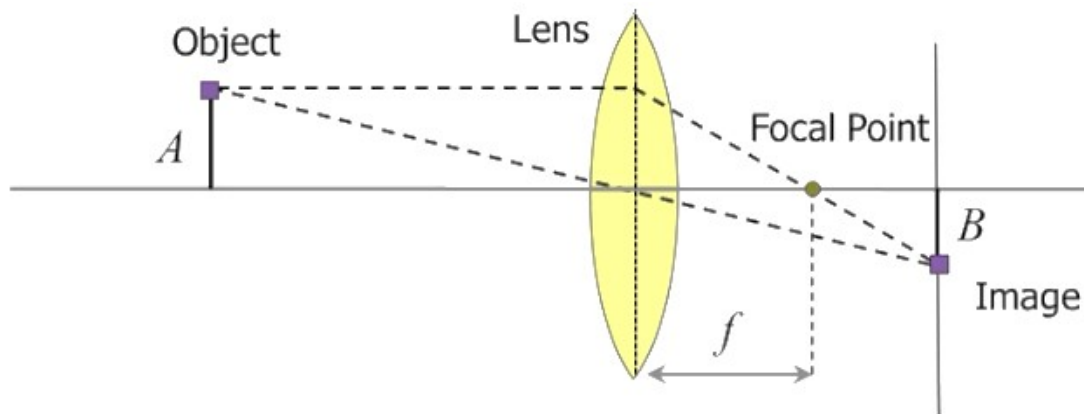
Mathematical Model of Imaging System



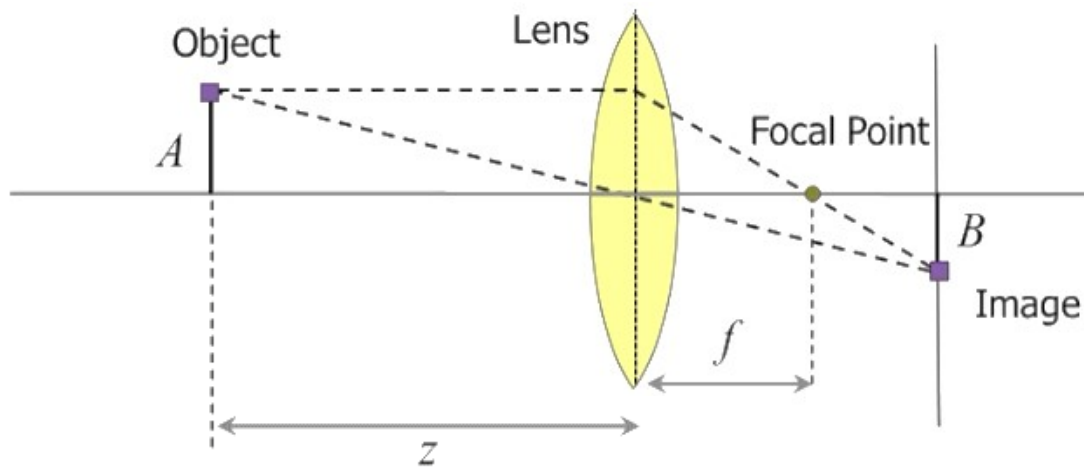
Mathematical Model of Imaging System



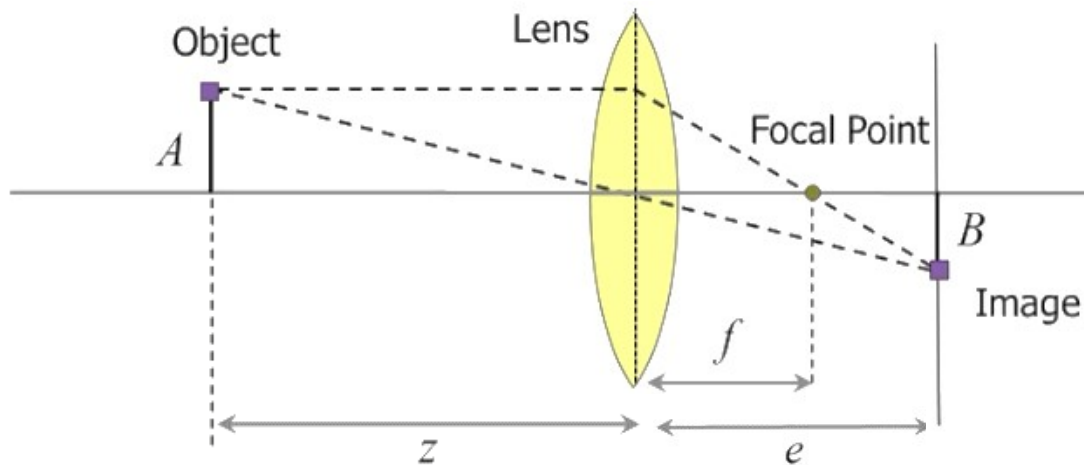
Mathematical Model of Imaging System



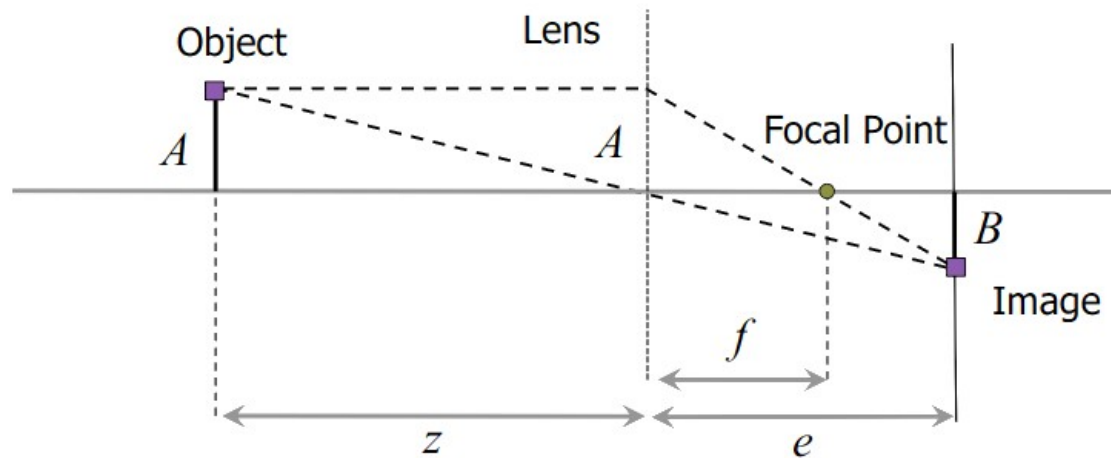
Mathematical Model of Imaging System



Mathematical Model of Imaging System

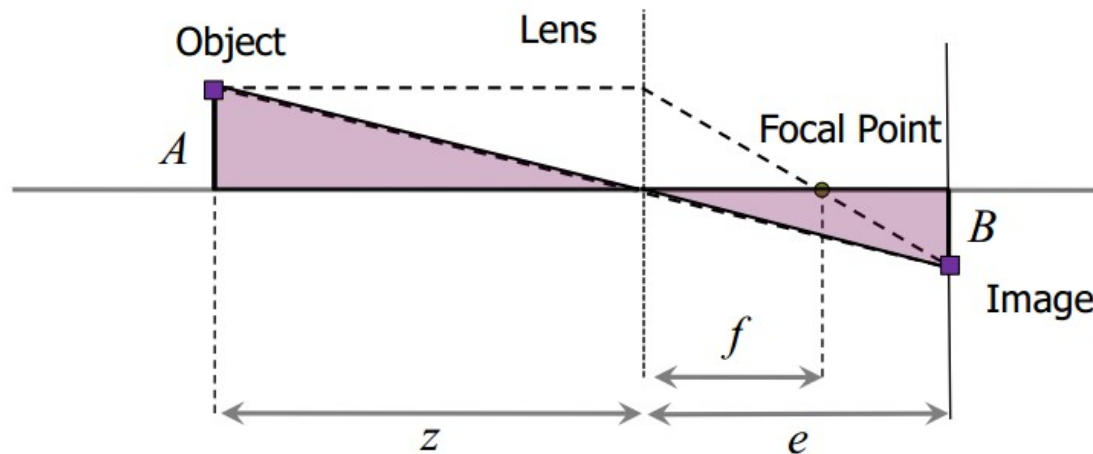


Mathematical Model of Imaging System



➤ What is the relationship between f , z , and e ?

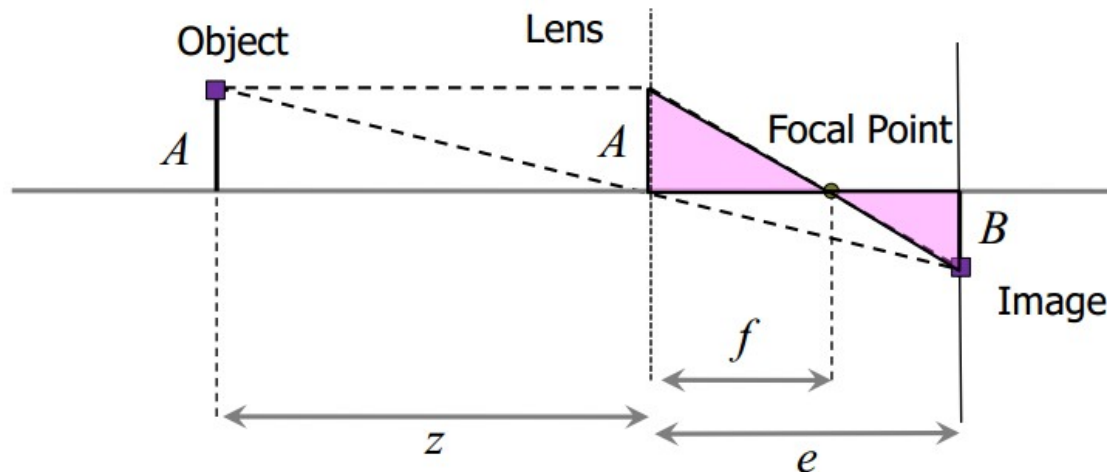
Mathematical Model of Imaging System



➤ What is the relationship between f , z , and e ?

➤ Principle of Similar Triangles
$$\frac{B}{A} = \frac{e}{z}$$

Mathematical Model of Imaging System

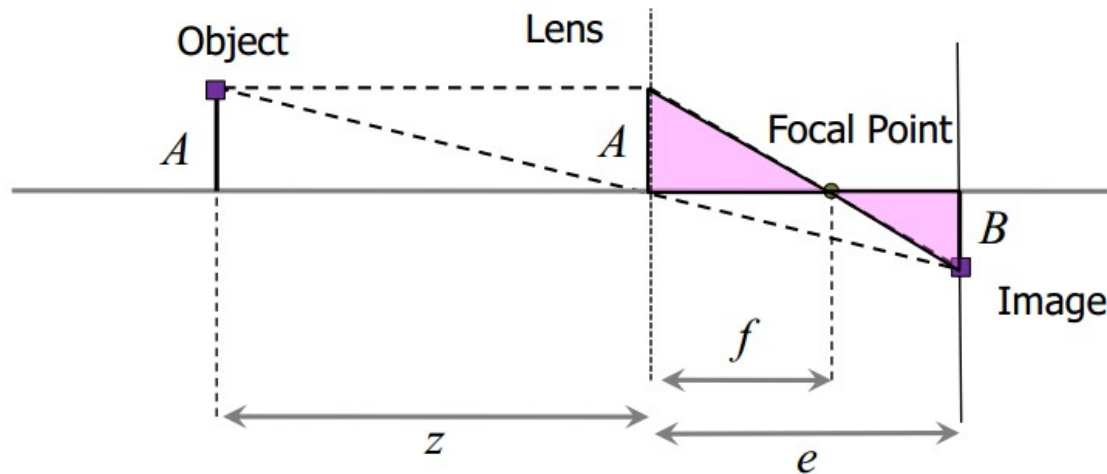


➤ What is the relationship between f , z , and e ?

➤ Principle of Similar Triangles

$$\frac{B}{A} = \frac{e}{z} \quad \frac{B}{A} = \frac{e-f}{f} = \frac{e}{f} - 1$$

Mathematical Model of Imaging System



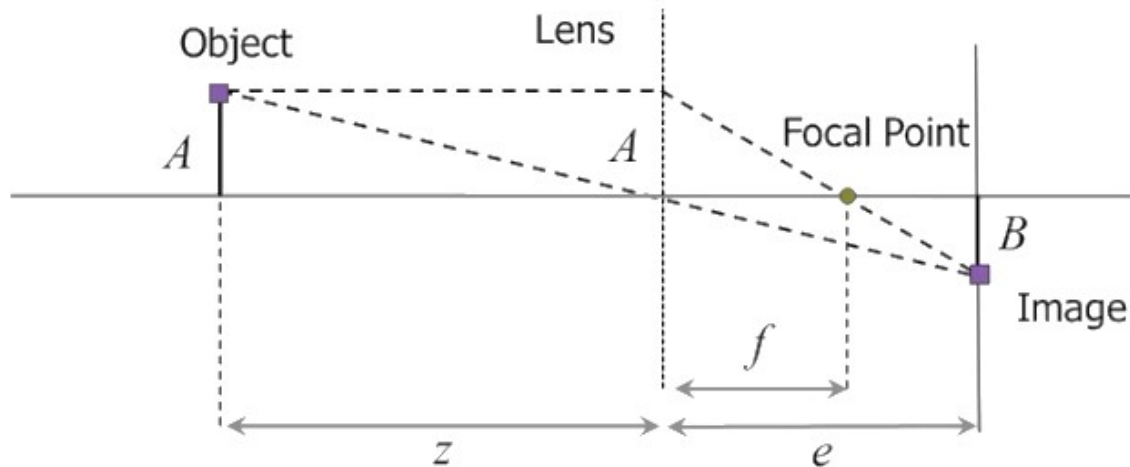
➤ What is the relationship between f , z , and e ?

➤ Thin lens formula

$$\frac{e}{f} - 1 = \frac{e}{z} \Rightarrow \boxed{\frac{1}{f} = \frac{1}{z} + \frac{1}{e}}$$

Mathematical Model of Imaging System

- What happens if $z \gg f$?

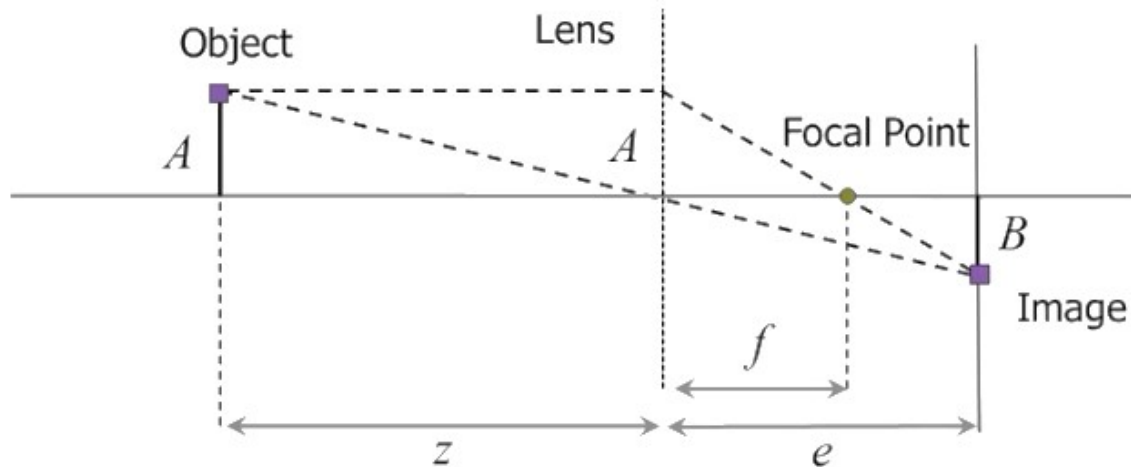


- The image plane needs to be adjusted to focus objects at infinity.

$$\frac{1}{f} = \frac{1}{z} + \frac{1}{e}$$

Mathematical Model of Imaging System

- What happens if $z \gg f$?



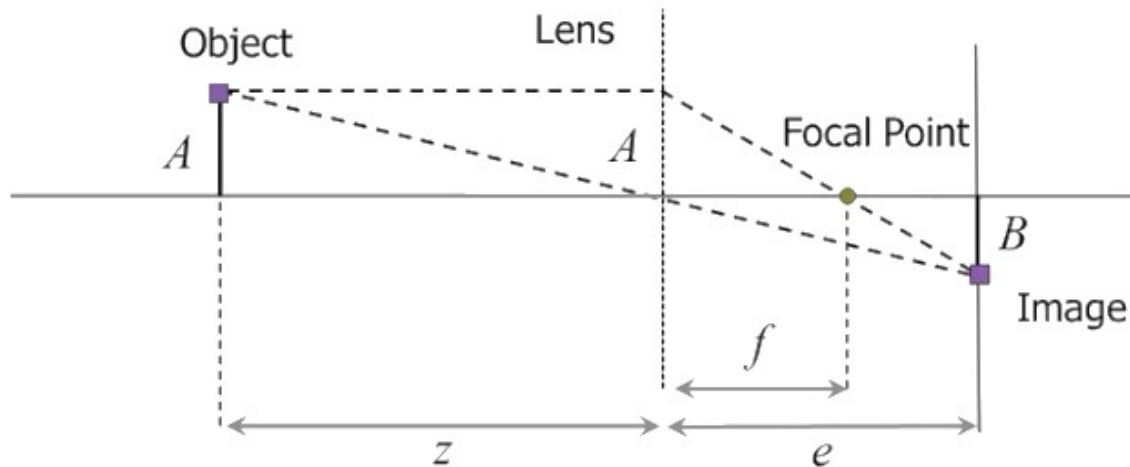
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$$\frac{1}{f} = \frac{1}{z} + \frac{1}{e}$$

$\frac{1}{z} \cong 0$

Mathematical Model of Imaging System

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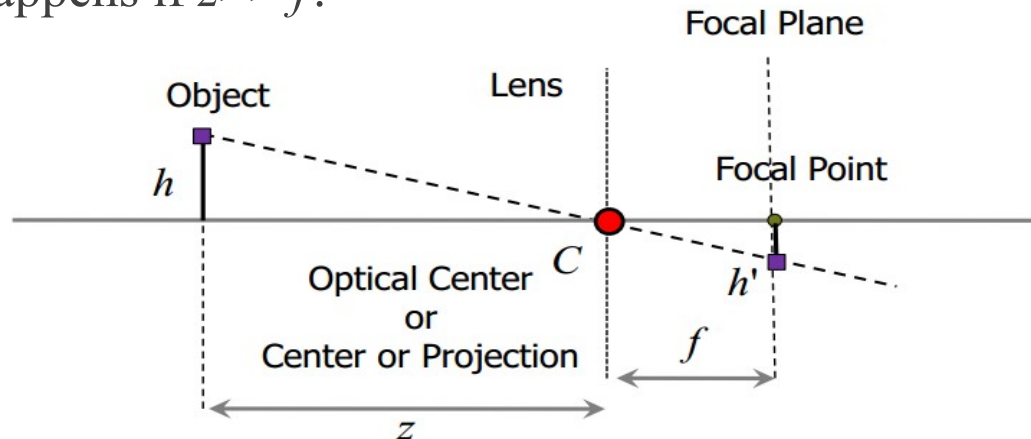


- The image plane needs to be adjusted to focus objects at infinity.

$$\frac{1}{f} = \underbrace{\frac{1}{z}}_{\cong 0} + \frac{1}{e} \Rightarrow \frac{1}{f} \approx \frac{1}{e} \Rightarrow f \approx e$$

Mathematical Model of Imaging System

➤ What happens if $z \gg f$?

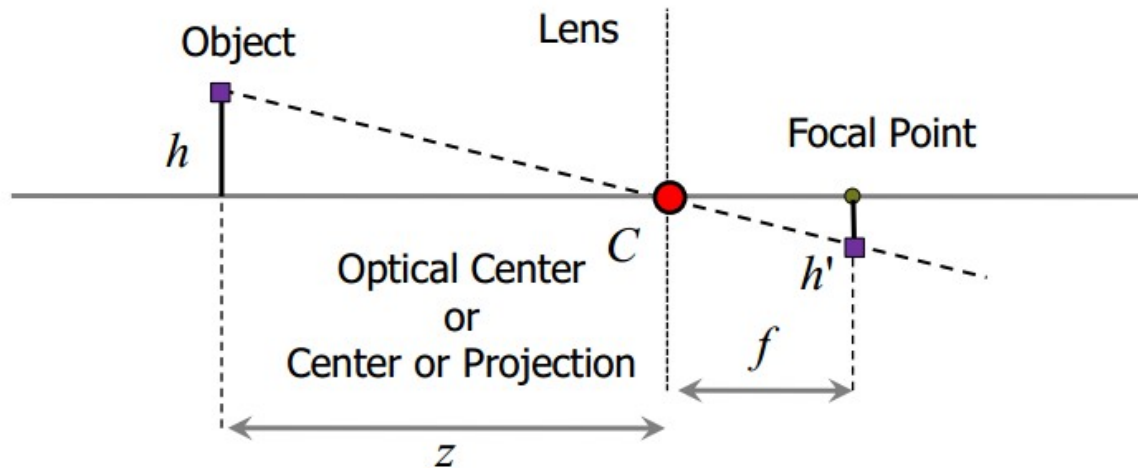


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Mathematical Model of Imaging System

➤ What happens if $z \gg f$?



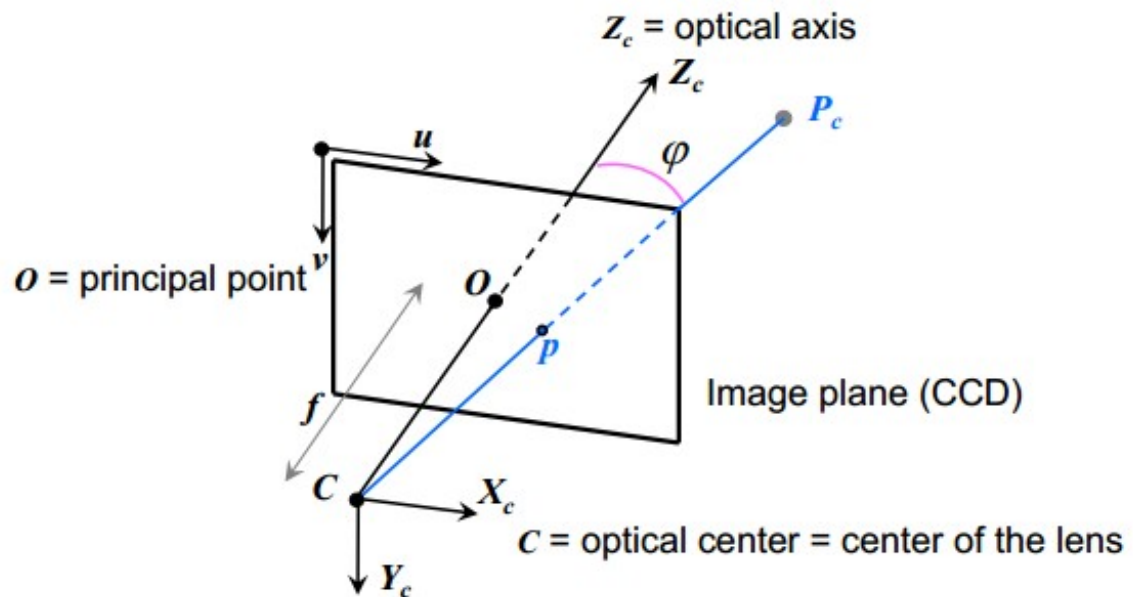
$$\frac{h'}{h} = \frac{f}{z} \Rightarrow h' = \frac{f}{z} h$$

Mathematical Model of Imaging System

- For convenience, the image plane is typically represented in front of the optical center so that the image maintains the same orientation (i.e., without flipping).

Mathematical Model of Imaging System

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- **A camera does not measure distance, but rather angles!**



Mathematical Model of Imaging System

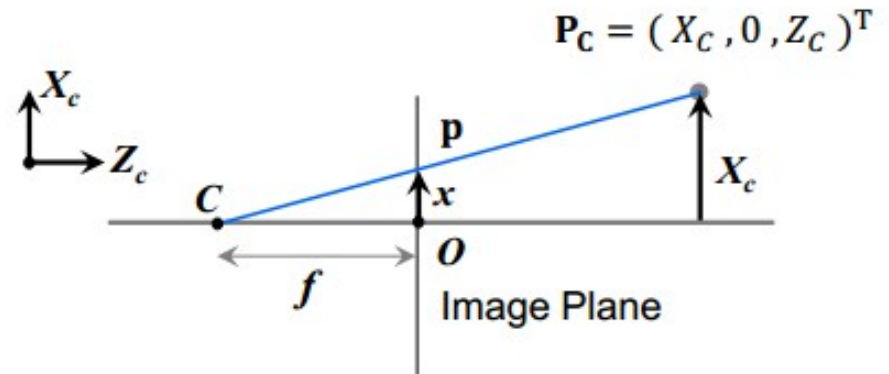
➤ In the camera coordinate system, $P_c = (X_c, 0, Z_c)^T$ is projected onto the image plane to form $p = (x, y)$.

➤ By similar triangles :

$$\frac{x}{f} = \frac{X_c}{Z_c} \Rightarrow x = \frac{fX_c}{Z_c}$$

➤ Similarly :

$$\frac{y}{f} = \frac{Y_c}{Z_c} \Rightarrow y = \frac{fY_c}{Z_c}$$



Mathematical Model of Imaging System

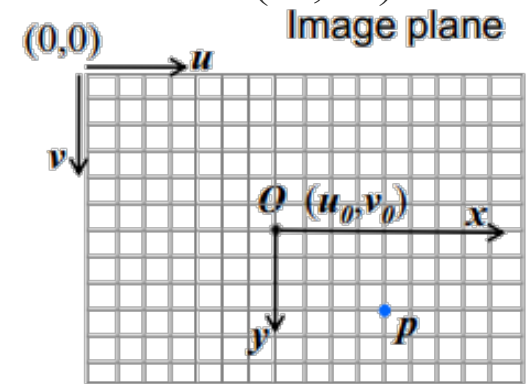
➤ To convert p from local image plane coordinates (x, y) to pixel coordinates (u, v) , consider:

- The pixel coordinates of the camera's optical center $O = (u_0, v_0)$

- The scale factor k representing pixel size

$$u = u_0 + kx \Rightarrow u_0 + k \frac{fX_C}{z_C}$$

$$v = v_0 + ky \Rightarrow v_0 + k \frac{fY_C}{z_C}$$



➤ By introducing an additional element (scale), homogeneous coordinates are used to represent the linear mapping from 3D to 2D:

Mathematical Model of Imaging System

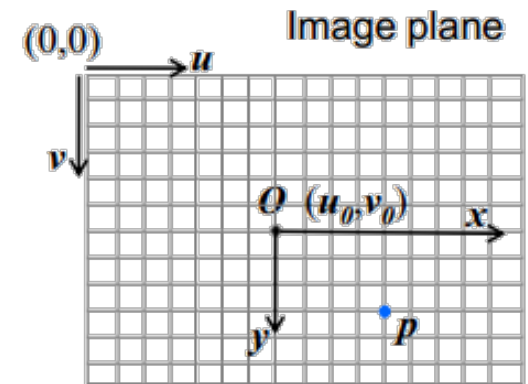
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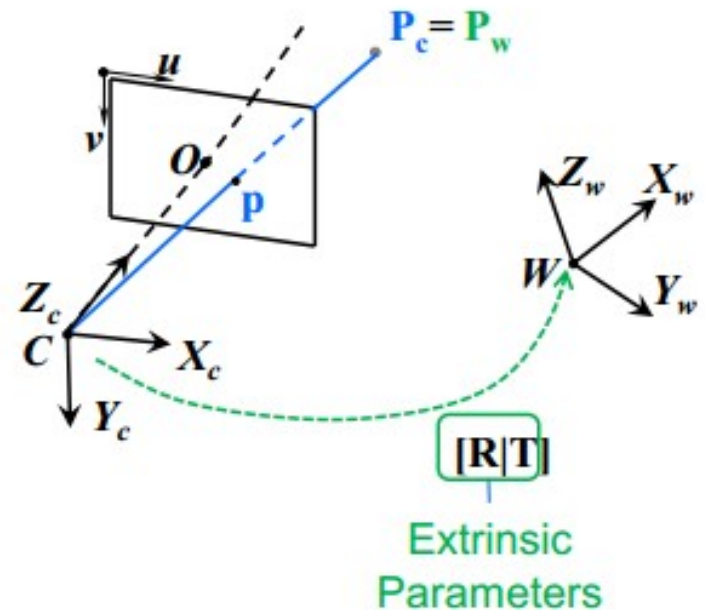
$$v = v_0 + ky \Rightarrow v_0 + k \frac{fY_C}{Z_C}$$



➤ By introducing an additional element (scale), homogeneous coordinates are used to represent the linear mapping from 3D to 2D:

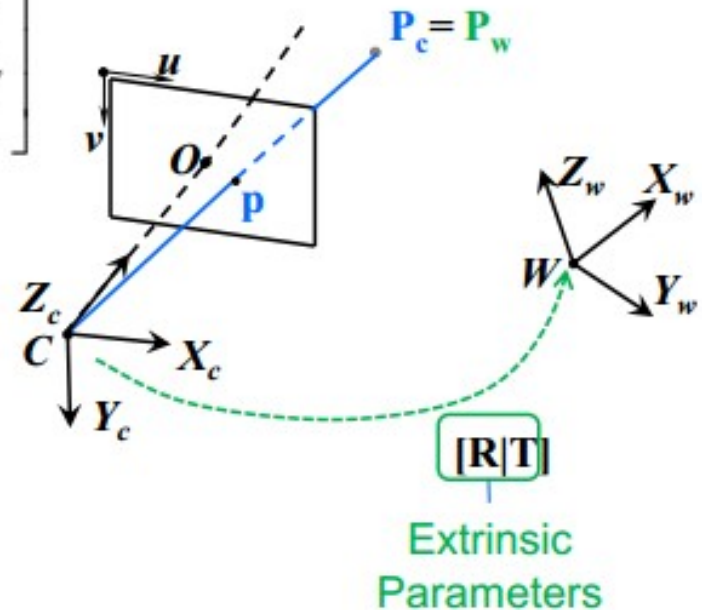
$$p = \begin{pmatrix} u \\ v \end{pmatrix} \Rightarrow \tilde{p} = \begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} = \lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}$$

Mathematical Model of Imaging System



Mathematical Model of Imaging System

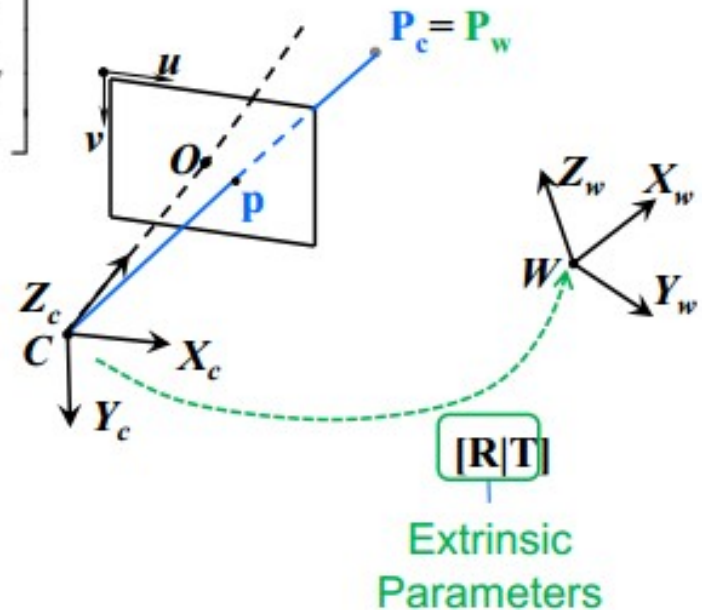
$$\begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} + \begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix} = \begin{bmatrix} R & | & T \end{bmatrix} \cdot \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$$



Mathematical Model of Imaging System

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$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix}$$



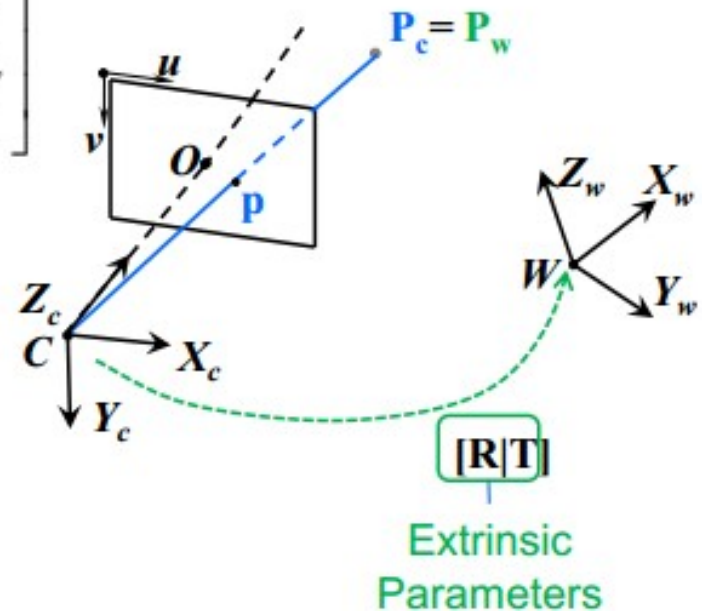
Mathematical Model of Imaging System

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$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix}$$

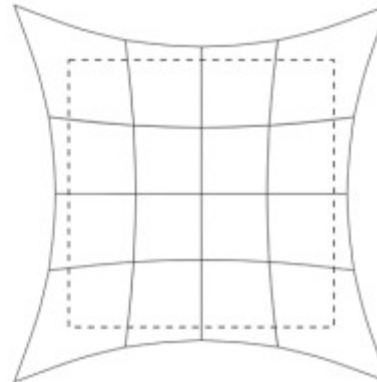
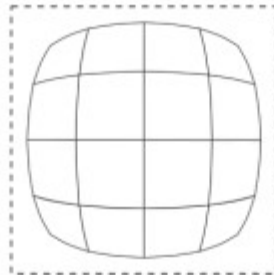
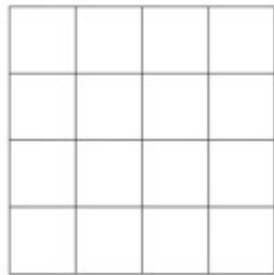
Perspective Projection Matrix

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \boxed{[R|T]} \cdot \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$$



Mathematical Model of Imaging System

➤ Distortion Model



No distortion



Barrel distortion



Pincushion

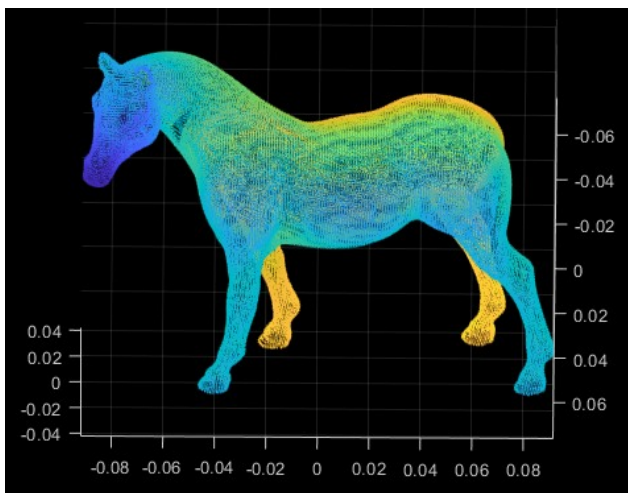
Mathematical Model of Imaging System

- The standard model for radial distortion is a transformation from ideal coordinates (u, v) , i.e., undistorted, to real observable coordinates (distorted) (u_d, v_d) .
- In this case, the image coordinate distortion is a nonlinear function of its radial distance. For most lenses, a simple quadratic distortion model yields good results.

$$\begin{bmatrix} u_d \\ v_d \end{bmatrix} = (1 + k_1 r^2) \begin{bmatrix} u - u_0 \\ v - v_0 \end{bmatrix} + \begin{bmatrix} u_0 \\ v_0 \end{bmatrix}$$

where

$$r^2 = (u - u_0)^2 + (v - v_0)^2$$



```
%Load a 3D point cloud
ptcloud=pcread('h:\data_sets\3D\PCDdata\horse.ply');
gammaobj= [0, pi/2, -pi/2];
Rxobj= [1 0 0; 0 cos(gammaobj(1)) sin(gammaobj(1)); 0 -sin(gammaobj(1)) cos(gammaobj(1))];
Ryobj= [cos(gammaobj(2)) 0 -sin(gammaobj(2)); 0 1 0; sin(gammaobj(2)) 0 cos(gammaobj(2))];
Rzobj= [cos(gammaobj(3)) sin(gammaobj(3)) 0; -sin(gammaobj(3)) cos(gammaobj(3)) 0; 0 0 1];
Robj=Rzobj*Ryobj*Rxobj;

ptnum=length(ptcloud.Location(:,1));
ptcloudxyz= (Robj*ptcloud.Location')';
ptcloud=pointCloud(ptcloudxyz);
```

```
figure,
```

```
%Define
```

```
f=8e-3;
```

```
v0=imag
```

```
%The in
```

```
K=[f/du
```

```
0,
```

```
0,
```

```
%The ex
```

```
gamma=
```

```
Rx= [1
```

```
Ry= [co
```

```
Rz= [co
```

```
R=Rz*Ry
```

```
T=[0, 0
```

```
cam = p
```

```
%The im
```

```
img=zeros
```

```
for i=1
```

```
pixel=K*[R, T]*[ptcloud.Location(i, :), 1]';
```

```
pixel=round(pixel/pixel(3));
```

```
if pixel(1)>image_size(1) || pixel(2)>image_size(1) || pixel(1)<1 || pixel(2)<1
    continue;
end
```

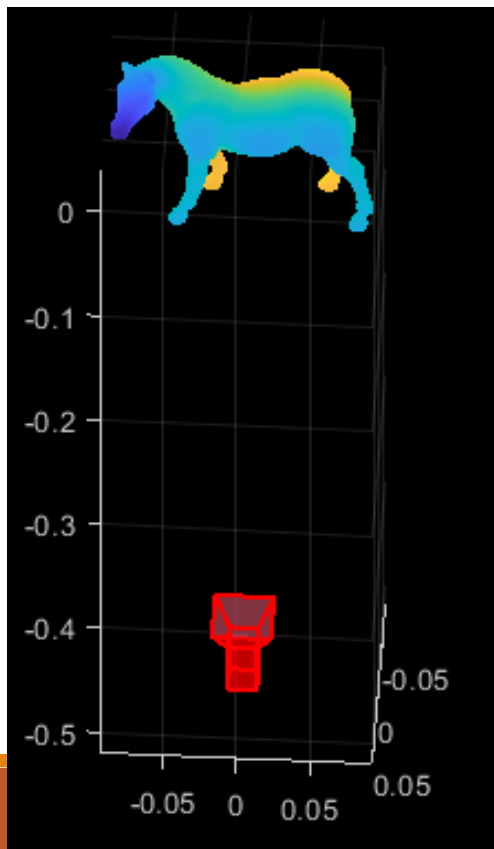
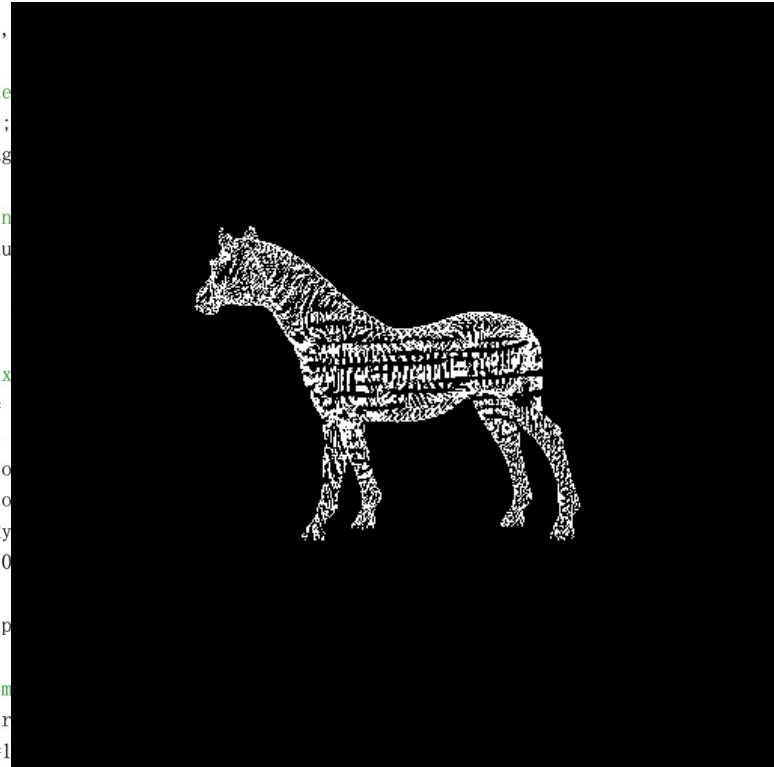
```
end
```

```
img(pixel(2), pixel(1))=1;
```

```
end
```

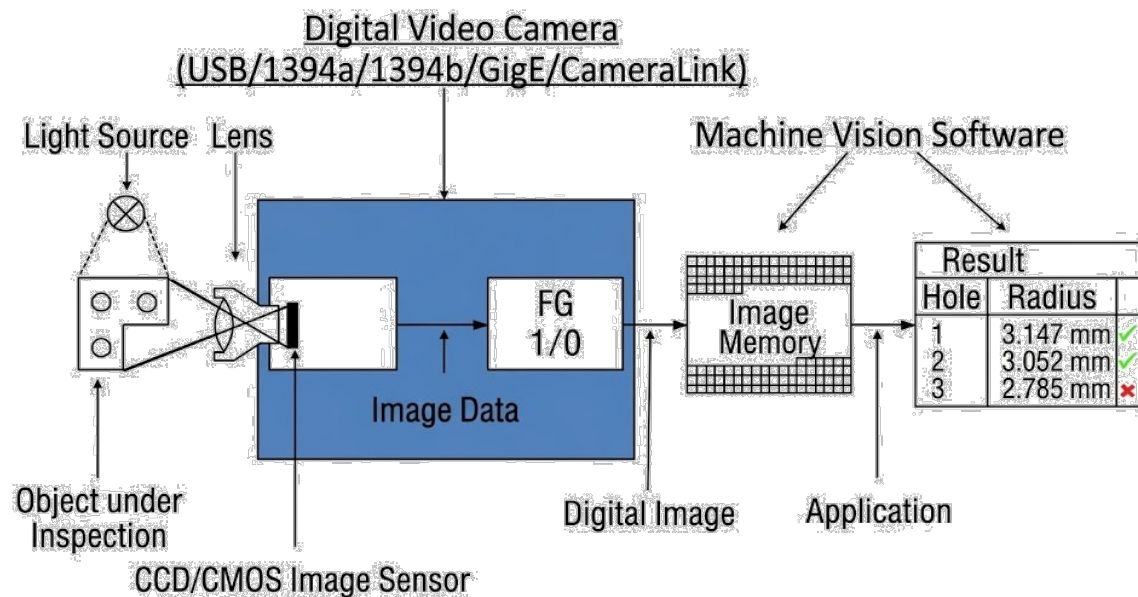
```
figure,
```

```
imshow(img);
```



Machine Vision Hardware Selection

➤ Many factors affect the imaging quality of machine vision systems, mainly including illumination, system resolution, pixel resolution, contrast, depth of field (DOF), perspective error, and lens distortion.



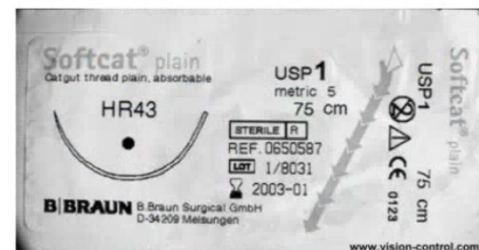
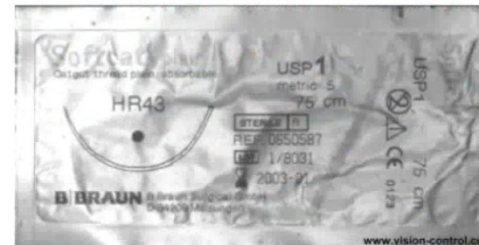
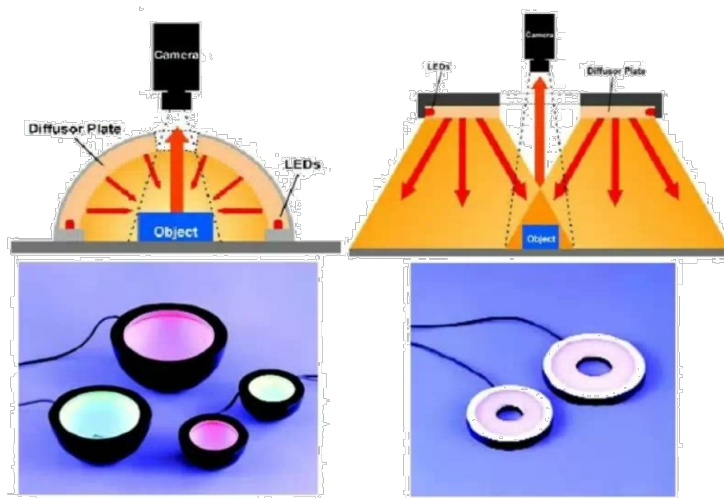
Machine Vision System Selection — Light Source

➤ Characteristics of LED Light Sources

- Can be made into various shapes, sizes, and irradiation angles;
- Can be made into various colors according to needs, and brightness can be adjusted at any time; through the heat dissipation device, the heat dissipation effect is better, and the light brightness is more stable;
- Long service life (about 30,000 hours, intermittent service life is longer);
- Fast response, can reach maximum brightness within 10 μ s or shorter time;
- Power supply have external trigger, can be controlled by computer, fast startup speed, can be used as a strobe light;
- LEDs with low operating costs and long life will demonstrate greater advantages in terms of comprehensive cost and performance;
- Can be customized according to customer needs.

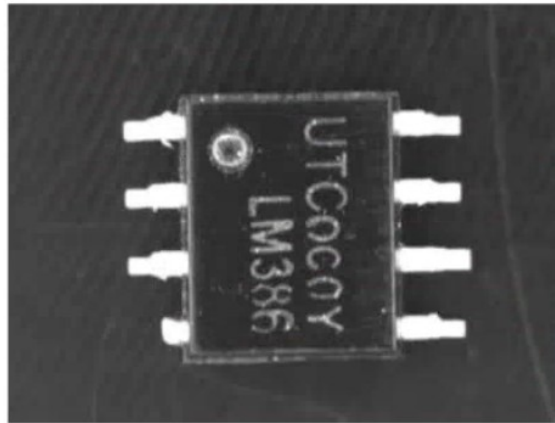
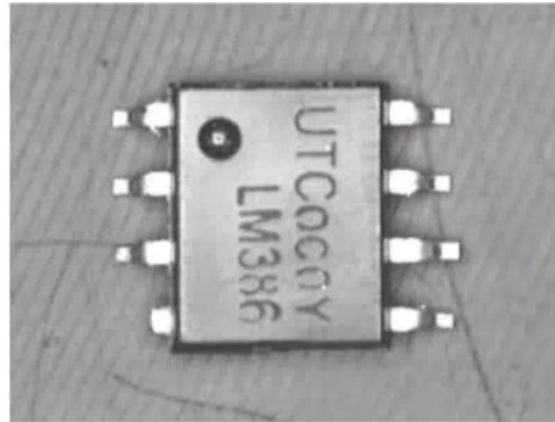
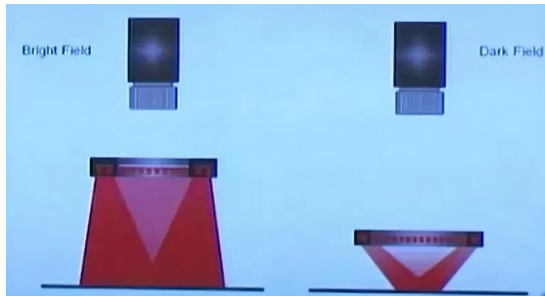
Machine Vision System Selection — Light Source

- In addition to varying in type, light sources also come in various shapes and designs, such as bar lights, ring lights, back lights (surface lights), and so on.
- General lighting typically employs ring-shaped or point-shaped illumination. Ring lights are a common general lighting method; they are easy to mount on lenses and can provide sufficient illumination for diffuse reflective surfaces.



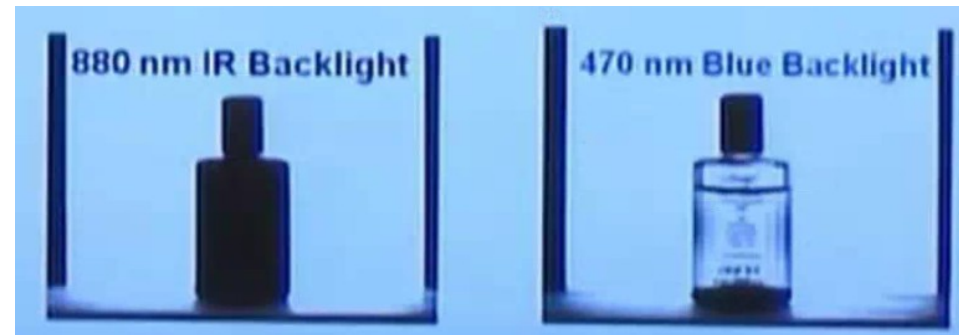
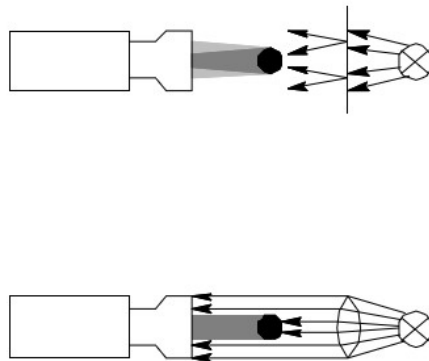
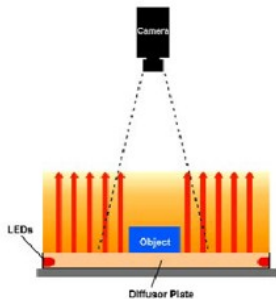
Machine Vision System Selection — Light Source

➤ Dark-field illumination refers to providing low-angle lighting relative to the object surface. When using a camera to capture an image within its field of view, if the light source is visible within that field, it is considered bright-field illumination; conversely, if the light source is not visible within the field of view, it is classified as dark-field illumination. Therefore, whether the illumination is bright-field or dark-field depends on the position of the light source. Typically, dark-field illumination is applied to highlight raised features on surfaces or variations in surface texture.



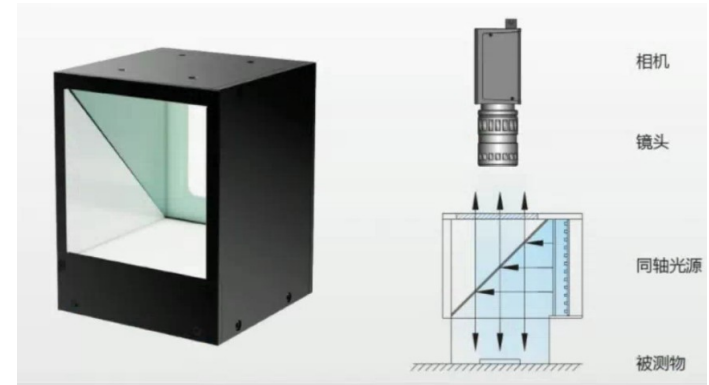
Machine Vision System Selection — Light Source

- Backlighting involves placing the light source behind the object relative to the camera. This lighting method differs greatly from other lighting methods because the image analyzes not reflected light but incident light.
- Backlighting produces very strong contrast. When applying backlighting technology,
- object surface features may be lost. For example, backlighting technology can be used to measure the diameter of a coin, but it is impossible to determine the front or back side of the coin.



Machine Vision System Selection — Light Source

- Coaxial lighting involves illuminating the surface of an object with light directed along the same axis as the camera. Coaxial lighting uses a special semi-reflective mirror to reflect the light source towards the camera's lens axis. The semi-reflective mirror only allows light reflected from the object's surface perpendicular to the lens to pass through.
- Coaxial lighting technology is very useful for achieving uniform illumination on flat objects with specular surfaces. Additionally, this technique can highlight areas where the surface angle changes, as light reflected from surfaces not perpendicular to the camera lens will not enter the lens, resulting in a darker appearance.

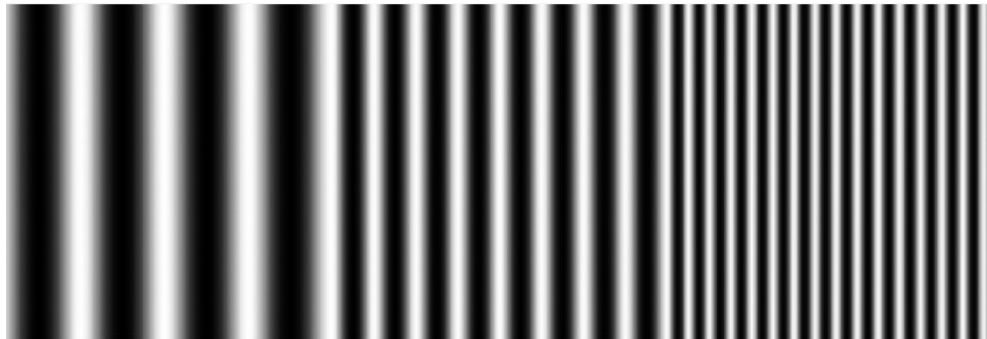


Machine Vision System Selection — Lens

- The lenses used in machine vision imaging systems are typically designed by combining convex and concave lenses.
- Convex lenses can converge light and are also referred to as converging lenses or positive lenses. Concave lenses have a diverging effect on light and are also referred to as diverging lenses or negative lenses.
- The main technical parameters related to lenses include lens resolution, focal length, minimum working distance, maximum image plane, field of view/angle of view, depth of field, aperture, relative aperture, and mounting interface type, etc.

Machine Vision System Selection — Lens

- Lens resolution represents its spatial limit resolving capability, commonly tested by imaging sinusoidal gratings.
- The higher the lens resolution, the greater the number of line pairs it can resolve per millimeter. For machine vision system design, the lens resolution can be obtained directly from the lens specification table.
- In contrast, a camera's spatial resolution also denotes its spatial limit resolving capability and can be calculated based on the physical size of its pixels.



Machine Vision System Selection — Lens

- For an imaging system composed of a lens and a camera, the spatial resolution of the entire system is determined by the minimum of the spatial resolutions of the lens and the camera.
- Only when the spatial resolution of the lens is greater than that of the camera can the optimal spatial resolution of the imaging system be ensured.
- In machine vision system design, the commonly referred system resolution does not refer to the system's limiting spatial resolution for line pairs, but rather to the system's ability to identify the smallest features in the detection target.

$$l_{\min} = \frac{\text{FOV}}{R_{\min}} \times p_{\min} \quad (p_{\min} \geq 2)$$

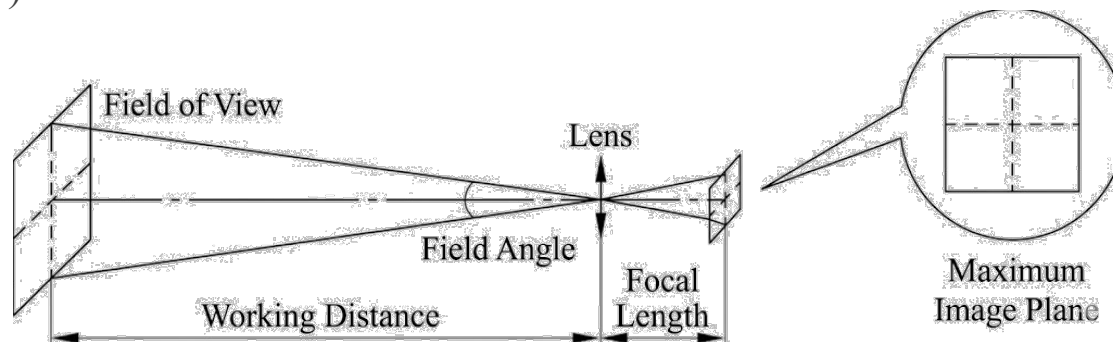
Machine Vision System Selection — Lens

- Focal length refers to the distance between the image position formed by an object at infinity on the image side of the lens and the image-side principal plane.
- Focal length embodies the basic characteristics of a lens: namely, at different object distances, the image position and image size of the target are determined by the focal length.
- Common focal lengths available on the market include 6mm, 8mm, 12.5mm, 25mm, and 50mm, among others.
- Generally, the minimum working distance is specified in the lens product parameters. When the camera operates at a distance less than this minimum working distance, image distortion will occur, affecting the reliability of the machine vision system.



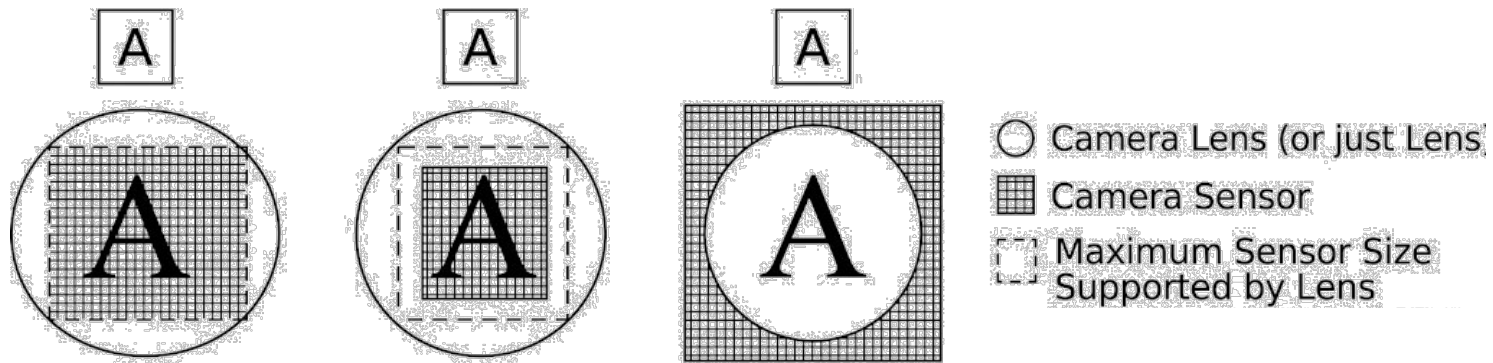
Machine Vision System Selection — Lens

- The maximum image plane refers to the largest clear imaging area that a lens can support (commonly expressed as the diameter of the observable range). Beyond this range, image contrast decreases and the image becomes blurred.
- The field of view of a lens is the observation area corresponding to its maximum image plane. The field of view angle is an alternative expression of the field of view and equals the angular subtense of the target corresponding to the maximum image plane.
- Typically, in long-distance imaging systems, such as telescopes and aerial photography lenses, the imaging range of a lens is measured by the field of view angle. In close-range imaging, it is commonly expressed by the actual diameter of the object plane (i.e., the format size).



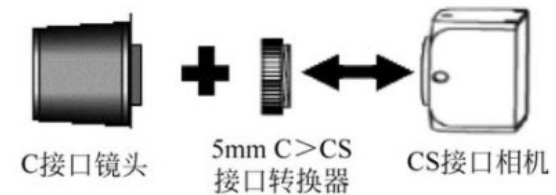
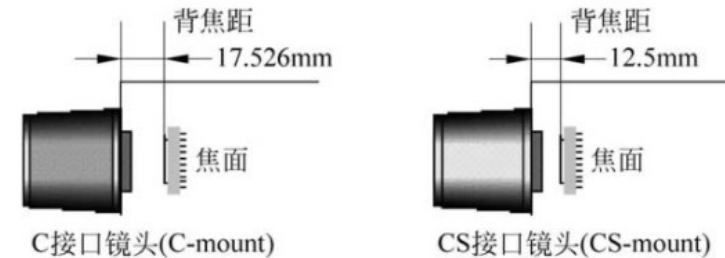
Machine Vision System Selection — Lens

- Lenses are often classified according to their field of view angle, such as telescopic lenses ($6^{\circ}\sim 12^{\circ}$), long-distance camera lenses ($12^{\circ}\sim 46^{\circ}$), standard lenses ($46^{\circ}\sim 65^{\circ}$), wide-angle lenses ($65^{\circ}\sim 100^{\circ}$), etc.
- When building machine vision systems (especially precision measurement systems), lenses with a focal length less than 8mm or a very large field of view angle are generally not selected.
- Generally, it must be ensured that the maximum sensor size supported by the selected lens is greater than or equal to the sensor size of the camera. Another main reason for doing this is to avoid vignetting.



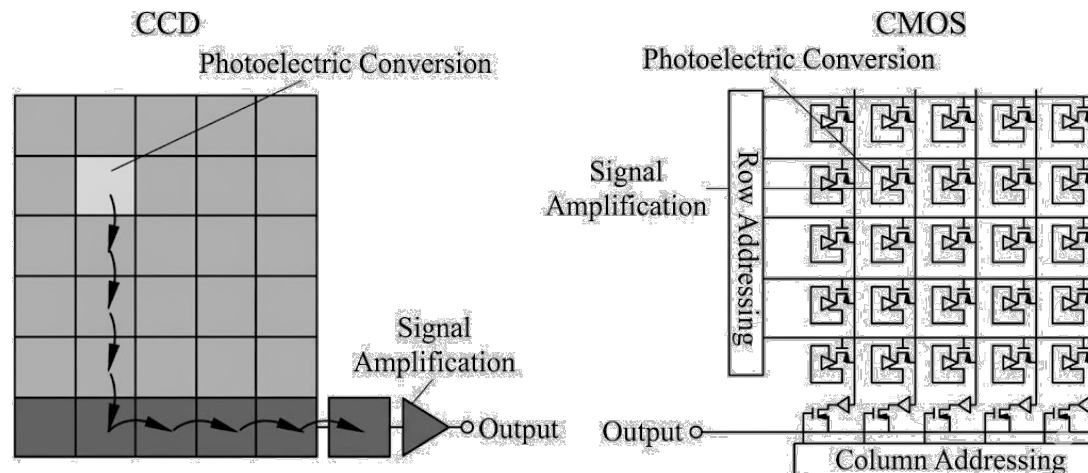
Machine Vision System Selection — Lens

- Common interface standards include the C-mount, CS-mount, and F-mount.
- Currently, lenses and cameras with C and CS mounts dominate the market, while F-mounts are typically used in high-resolution line-scan cameras (over 2048 pixels) to capture larger images than those achievable with C or CS mount lenses.
- A C-mount lens can be converted into a CS-mount lens by adding a 5mm extension tube (adapter); however, CS-mount lenses cannot be used with C-mount cameras.
- Additionally, 3CCD cameras with C-mounts are not compatible with C-mount lenses, so it is necessary to consult the lens compatibility list provided by the camera manufacturer.



Machine Vision System Selection – Imaging Chips

- The development of CMOS (Complementary Metal Oxide Semiconductor) image sensors first emerged in the early 1970s.
- In the early 1990s, with the development of Very Large Scale Integration (VLSI) manufacturing process technology, CMOS image sensors underwent rapid development.
- The photoelectric conversion principle of CMOS image sensors is the same as that of CCD image sensors; the main difference between the two lies in the method of charge transfer.



Machine Vision System Selection – Imaging Chips

- CCD sensor chips convert charge into analog signals, which are then amplified and undergo A/D conversion before being output as digital signals.
- CMOS sensor chips directly amplify each charge and convert it into a digital signal for output, often resulting in poor imaging consistency.
- During charge transfer, CCDs do not introduce distortion, and since signals are uniformly amplified before output, they offer high imaging quality and consistency.
- CCD sensors feature a larger fill factor and higher signal-to-noise ratio, making them more sensitive to light and better suited for low-contrast scenarios.
- CMOS sensors can achieve significantly higher image transmission speeds compared to CCD sensors, making them more suitable for high-speed applications.

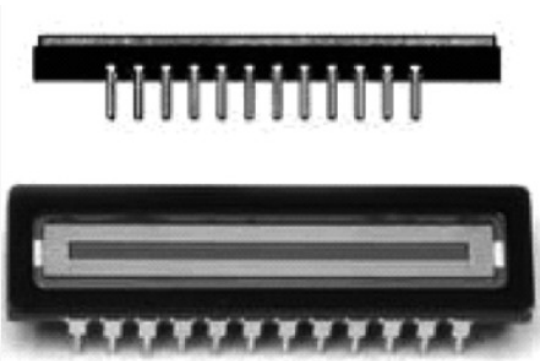
Machine Vision System Selection – Imaging Chips

- The signal from a CMOS sensor is amplified before transfer, so its power consumption is lower than that of a CCD, making it more suitable for portable devices.
- The CCD manufacturing process is relatively complex. Currently, only a few manufacturers such as Teledyne DALSA, Sony, and Panasonic master the core technology, so its price is generally higher.

Features	CCD	CMOS	Performance	CCD	CMOS
Pixel Output Signal	Charge	Voltage	Manufacturing Process	Complex	Simple
Chip Output Signal	Analog	Digital	Dynamic Range	High	Medium
Fill Factor	High	Medium	Pixel Uniformity	High	Low
Noise	Low	Medium-High	Light Sensitivity	High	Medium
Resolution	Low	High	Low Contrast Adaptation	Strong	Weak
Power Consumption	Medium-High	Low	Transmission Speed	Medium-High	High
Cost	Medium	Medium-Low			

Machine Vision System Selection – Imaging Chips

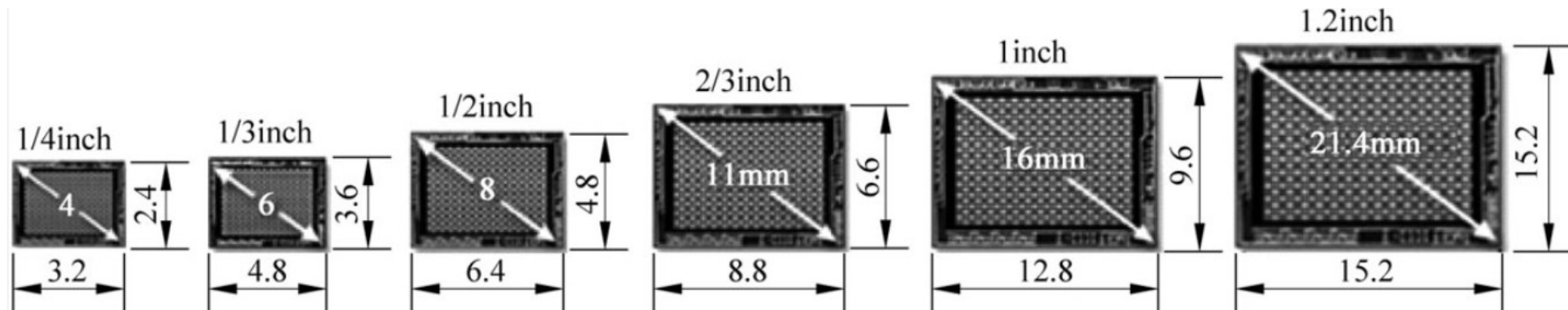
- The photosensitive area of a CCD image sensor can be arranged in a one-dimensional line array or a two-dimensional area array manner, and the corresponding cameras are line array (line scan) cameras or area array (area scan) cameras.
- Line array CCDs are typically composed of photosensitive units arranged one-dimensionally in numbers such as 1024 (1K), 2048 (2K), 4096 (4K), 6144 (6K), 12288 (12K), etc.
- Line array CCD sensors can also be classified into single-channel, dual-channel, four-channel, eight-channel CCD sensors, etc., based on the number of transmission channels. The greater the number of channels, the relatively faster the transmission speed.



1K	2K	4K	6K	12K
14mm				
22mm				
43mm				
62mm				
90mm				

Machine Vision System Selection – Imaging Chips

- In area CCD sensors, photosensitive elements are distributed in a two-dimensional array, and the size of the sensor chip is often represented by its diagonal length.
- Common sizes include 1 inch, 2/3 inch, 1/2 inch, 1/3 inch, and 1/4 inch, among others.
- Area CCDs can acquire a whole image at once, and according to the different methods of charge transfer, they can be classified into several types: Full Frame Transfer (FFT), Frame Transfer (FT), Interline Transfer (IT), and Frame Interline Transfer (FIT).



Machine Vision System Selection – Imaging Chips

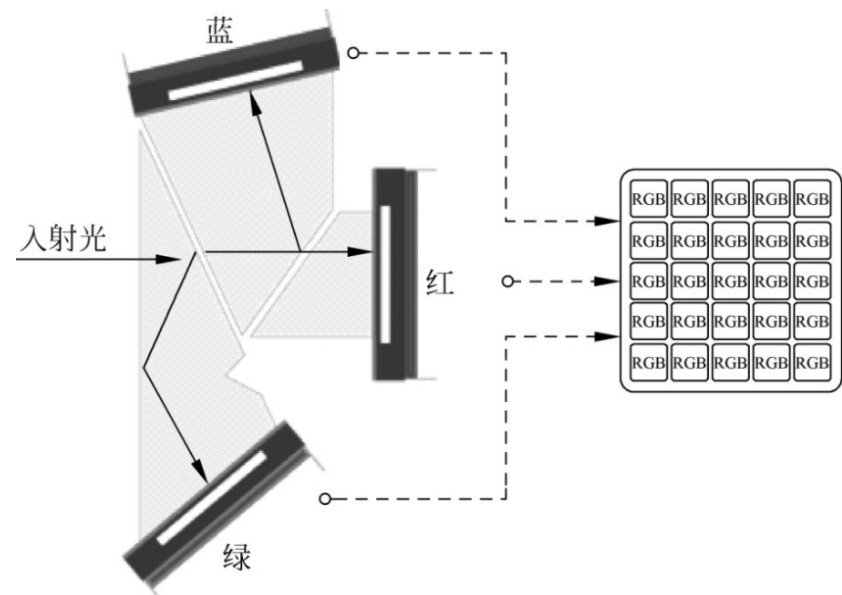
- The advantage of an area array CCD is that it can acquire two-dimensional image information, making the measurement image intuitive; the disadvantage is that although the total number of pixels is large, the number of pixels per row is generally fewer than that of a linear array, and the frame rate is limited.
- The advantage of a linear array CCD is that the number of one-dimensional pixels can be made very large, pixel sizes are relatively flexible, and the frame rate is high, making it particularly suitable for measuring one-dimensional dynamic targets. In situations requiring a large field of view and high image resolution, it cannot even be replaced by an area array CCD.
- However, to acquire a two-dimensional image using a linear array CCD, scanning motion must be employed. If the corresponding position of each pixel in the image on the measured object needs to be determined, devices such as gratings must also be equipped to record the coordinates of each scan line of the linear array CCD. This results in long image acquisition times for linear array CCDs, low measurement efficiency, and a relatively complex system. Moreover, image accuracy may be reduced due to the influence of scanning motion accuracy, ultimately affecting measurement accuracy.

Machine Vision System Selection – Imaging Chips

- Pixel resolution is one of the most critical parameters for CCD and CMOS sensors, defined by the number of pixels along the horizontal and vertical directions within the effective pixel area (i.e., the sensor size).
- CMOS cameras allow users to flexibly select regions of interest in the output image to enhance transmission speed. In contrast, CCD cameras can only increase transmission rates by reducing the number of rows, such as through interlaced output.
- Spectral response is another key parameter for CCD/CMOS sensors, indicating the sensor's sensitivity to light at different wavelengths.
- Typically, to align the spectral response of CCD/CMOS sensors with that of the human eye or to mitigate temperature-related effects on CCD cameras, an infrared cut filter (IR-cut filter) is installed at the factory. This filter blocks infrared light from passing through the camera lens, preventing image distortion.
- For applications requiring high spectral response to near-infrared light (around 750 nm), current standard CCD sensor technology can meet these requirements within a reasonable cost range.

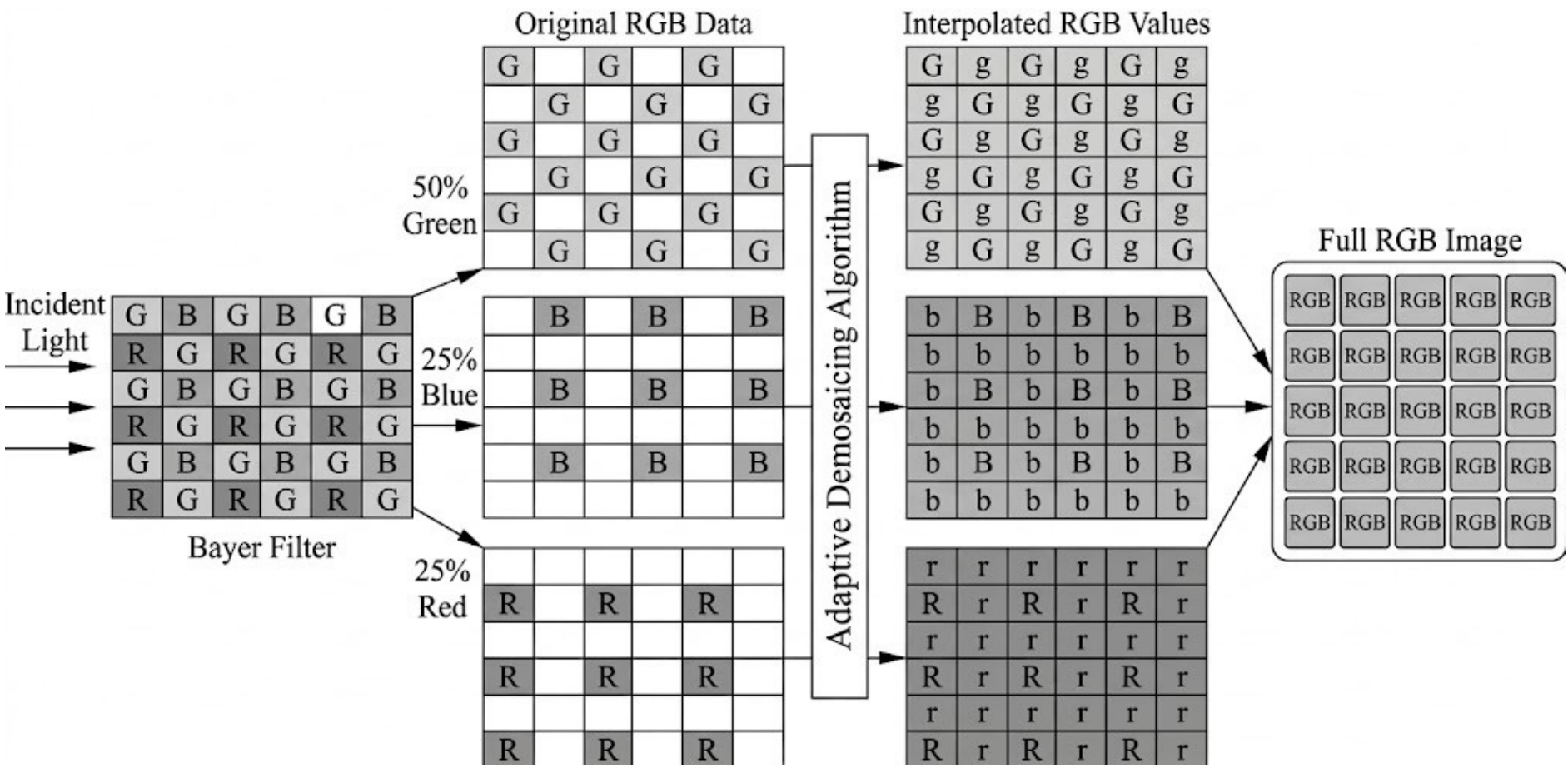
Machine Vision System Selection – Imaging Chips

- Essentially, CCD and CMOS sensors cannot distinguish colors on their own. When used as image sensors in color cameras, a dichroic prism can separate incoming light into red (R), green (G), and blue (B) components, which are then converted into electrical signals by three separate sensor chips.
- Cameras employing this structure are known as 3CCD-RGB color cameras.
- The advantage of 3CCD-RGB color cameras lies in their ability to accurately reproduce colors, making them suitable for applications requiring precise color measurement.
- However, due to their use of prisms and three sensor chips, they tend to be more expensive.



Machine Vision System Selection – Imaging Chips

- The Bayer filter adopts a mosaic-like color arrangement, where one row uses blue and green elements, and the next row uses red and green elements, alternating in this manner.
- The CCD chip beneath this filter has 2 out of every 4 pixels sensitive to the green component, 1 sensitive to blue, and 1 sensitive to red (there is no strict requirement for the arrangement order of pixels of different colors), thus making each physical pixel contain information of only one of the three colors: red (25%), green (50%), or blue (25%).
- When restoring the color information in the real image corresponding to each pixel, complete RGB distribution can be calculated based on the intensity information of color components within the neighborhood through a "color space interpolation algorithm".
- Color interpolation algorithms typically produce a low-pass effect, which causes images generated by single-CCD color cameras to contain significantly less detail than those generated by 3CCD color cameras.

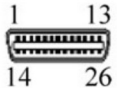
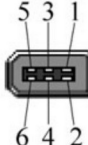
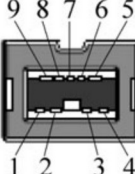
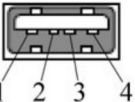
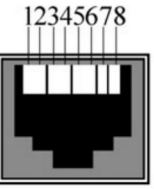


Machine Vision System Selection — Communication Protocol

- After CCD/CMOS chips complete photoelectric conversion, their output consists of analog or digital electrical signals.
- Typically, these signals undergo further amplification and correction, with synchronization, modulation, or sampling encoding added to generate video signals compliant with various standards before formal output.
- In theory, a camera's output signal could take any customized form. However, since television systems predate machine vision by many years, outputting signals in video formats already widely adopted in television systems not only facilitates signal transmission but also maximizes the utilization of existing mature hardware and software technologies.
- Except for a few cameras that output non-standard signals, most cameras produce either analog or digital video signals. Consequently, machine vision cameras are categorized into two main types based on their output signal formats: analog cameras and digital cameras.

Machine Vision System Selection — Communication Protocol

- Digital video signals mostly adopt progressive scanning instead of interlaced scanning, and use frame enable (frameenable) and line enable (line enable) signals to replace the field sync and line sync signals in analog video signals, to precisely control each line and each frame of the image.
- Currently, most digital cameras used in machine vision applications directly output RGB data.

接口标准	RS-422	LVDS (RS-644)	Camera Link	FireWire 1394. a	FireWire 1394. b	USB 2.0	GigE
连接器 示意图							
连接器	68/100pin	68/100pin	26pin	6pin	9pin	USB	RJ45
传输速率	10Mb/s	400Mb/s	最高 3.6Gb/s	400Mb/s	800Mb/s	480Mb/s	1000Mb/s
传输距离	1200m	最大 35m	7~10m	4.5m	100m (GOF 线缆)	3~7m	100m
设备数	10	1	1	最多 63	最多 63	最多 127	无限制
拓扑结构	到采集卡	到采集卡	点对点	点对点	点对点	主/从	点对点

Machine Vision System Selection — Hardware Selection

- Pixel (pixel): The recorded value corresponds to each point in a digital image.
- Resolution (image resolution): The number of pixels horizontally and vertically in a digital image.
- System resolution refers to the smallest detail or smallest feature of a monitoring target that an imaging system can identify.
- Pixel resolution refers to the number of pixels required to represent a detection target. If there are no special requirements, at least two pixels are commonly used to represent the smallest feature in the detection target.
- The minimum pixel resolution an image sensor should possess is often calculated using the following formula:

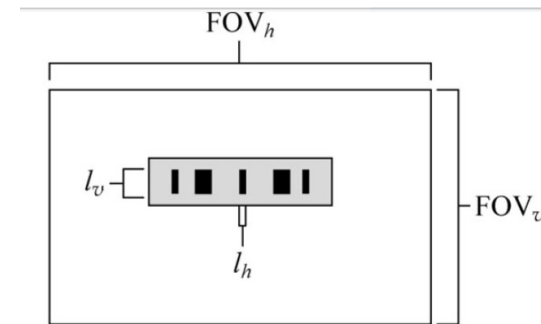
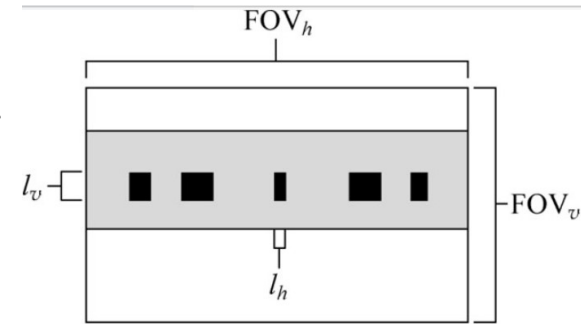
$$R_{\min} = \frac{L_{\max}}{l_{\min}} \times p_{\min}$$

Machine Vision System Selection — Hardware Selection

➤ Field of View (FOV) refers to the maximum area that an image sensor can monitor in an imaging system. When designing a machine vision system, considering that the detected target is generally made to fill the entire field of view as much as possible, the field of view size is often used instead of the target's maximum length L_{max} to calculate the pixel resolution of the vision system.

➤ If the field of view sizes in the horizontal and vertical directions are $[FOV_h, FOV_v]$, and the sizes of the minimum features of the detection target are $[l_h, l_v]$, then the minimum pixel resolution the image sensor should have is

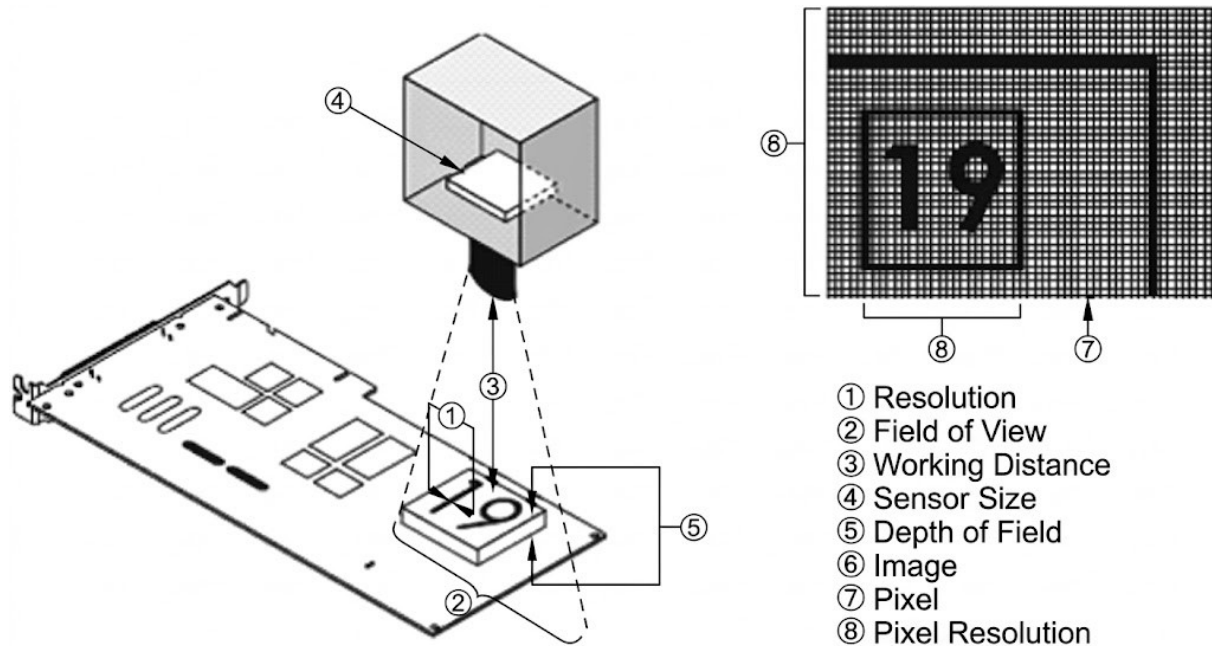
$$R_{\min} = \left[\frac{FOV_h}{l_h} \times p_{\min}, \frac{FOV_v}{l_v} \times p_{\min} \right]$$



Machine Vision System Selection — Hardware Selection

➤ Parameter Constraint Relationships of Imaging Systems Under the Pinhole Model:

$$\text{FOV} = \frac{S}{f} \times \text{WD} = M \cdot \text{WD} = \frac{R_{\min} \times l_{\min}}{p_{\min}} \quad (\text{Unless otherwise specified, } p_{\min} = 2)$$



Machine Vision System Selection — Hardware Selection

- After the lens and camera selection is completed, the working distance should be adjusted within the range of the maximum field of view and maximum installation distance defined by the project requirements, so that the field of view when the camera is working is

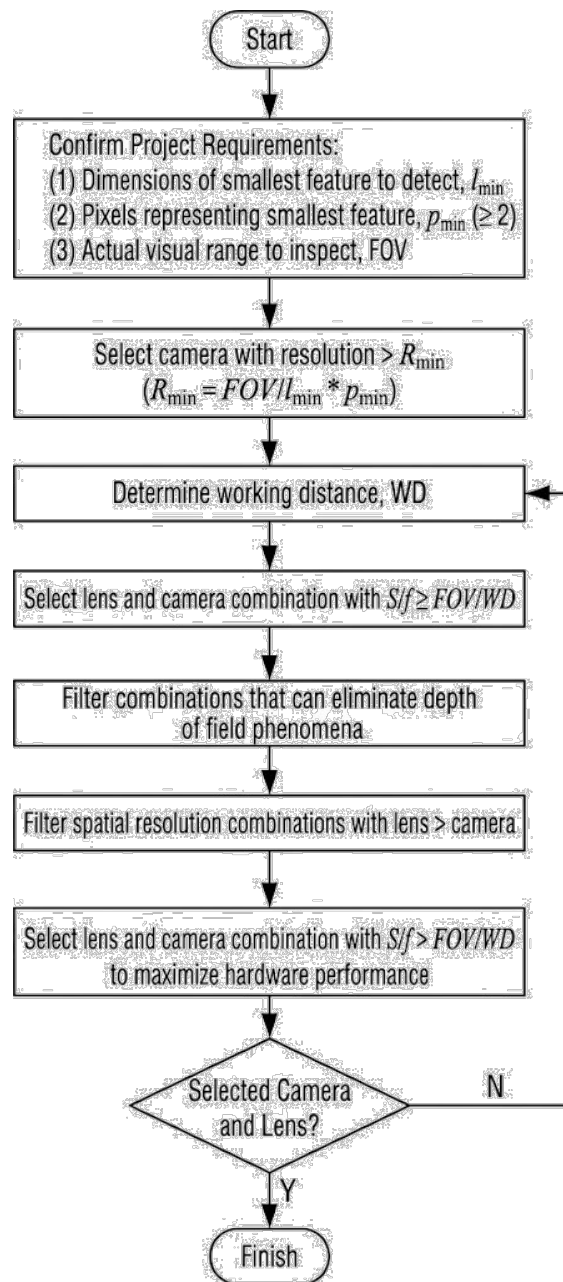
$$\text{FOV} = L_{\text{object}} + l_{\text{dynamic}} + l_{\text{buffer}}$$

- Where, L_{object} is the maximum length of the detection target, l_{dynamic} is the maximum distance the target may move, l_{buffer} is the reserved safety distance, Usually equals $0.1 \times L_{\text{object}}$, thus ensuring that on the premise of guaranteeing the detection target is captured by the camera, there are more pixels to represent the minimum feature of the target.
- If neither the camera nor the lens is determined in advance, it is necessary to first understand the requirements of the project working environment regarding camera installation (working distance), the maximum range to be detected (field of view), the size of the minimum feature and the number of pixels representing it, and then calculate what kind of lens or camera should be used based on these conditions.

Machine Vision System Selection — Hardware Selection

➤ Selecting a camera for a machine vision system is a step-by-step filtering process that comprehensively considers the lens, the interface between the image acquisition card and the camera, as well as the camera's own parameters. This can primarily be approached from the following aspects:

- First, determine the light source, lens, sensor size, and resolution based on project requirements and the machine vision imaging system model.
- Second, determine the camera's output method and standards (analog/digital, color, rate, etc.).
- Third, determine the camera's physical and electrical interfaces. Fourth, determine other camera performance indicators.



Parameter Constraint Relationship of the Simplified Imaging System Model:

$$\frac{S}{f} \times WD = FOV = \frac{R_{min} \times l_{min}}{p_{min}}$$

l_{min} : Dimensions of smallest feature to detect

p_{min} : Pixels representing smallest feature

FOV : Actual visual range to inspect

FOV_{max} : Maximum field of view

R_{min} : Minimum camera resolution

WD : Working distance

S : Sensor size

f : Lens focal length

Camera Calibration



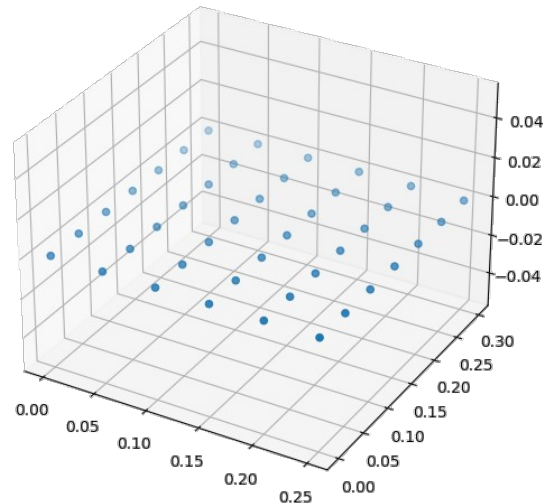
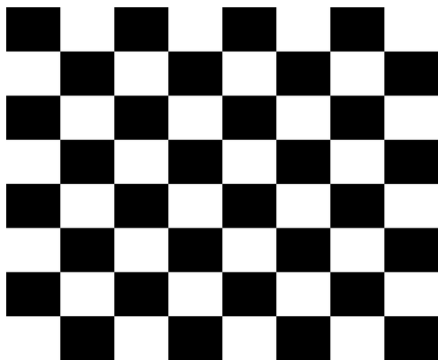
➤ Camera calibration aims to address:

- The internal parameters of the camera. This includes the focal length, optical center, and distortion model of each camera.
- The external parameters of the camera. This includes the rotation and translation relationships between each camera coordinate system and the world coordinate system.
- The relative positional relationships between cameras. This includes the rotation and translation relationships between each camera coordinate system and other camera coordinate systems, as well as the coordinates of spatial points corresponding to homologous image points in multiple images.

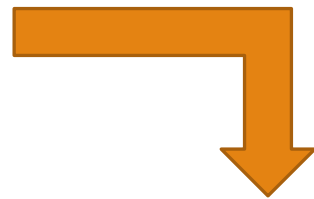
Camera Calibration



Zhang's Camera Calibration Method



$$s \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = A[R | \mathbf{t}] \begin{pmatrix} X \\ Y \\ 0 \\ 1 \end{pmatrix} = A[\mathbf{r}_1 | \mathbf{r}_2 | \mathbf{t}] \begin{pmatrix} X \\ Y \\ 1 \end{pmatrix}$$



$$\tilde{\mathbf{m}} = H\tilde{\mathbf{M}} \quad \text{with} \quad H = A[\mathbf{r}_1 \quad \mathbf{r}_2 \quad \mathbf{t}]$$

Zhang's Camera Calibration Method

Express the correspondence between H and A , R as:

$$\begin{bmatrix} \mathbf{h}_1 & \mathbf{h}_2 & \mathbf{h}_3 \end{bmatrix} = \lambda A \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{t} \end{bmatrix}$$

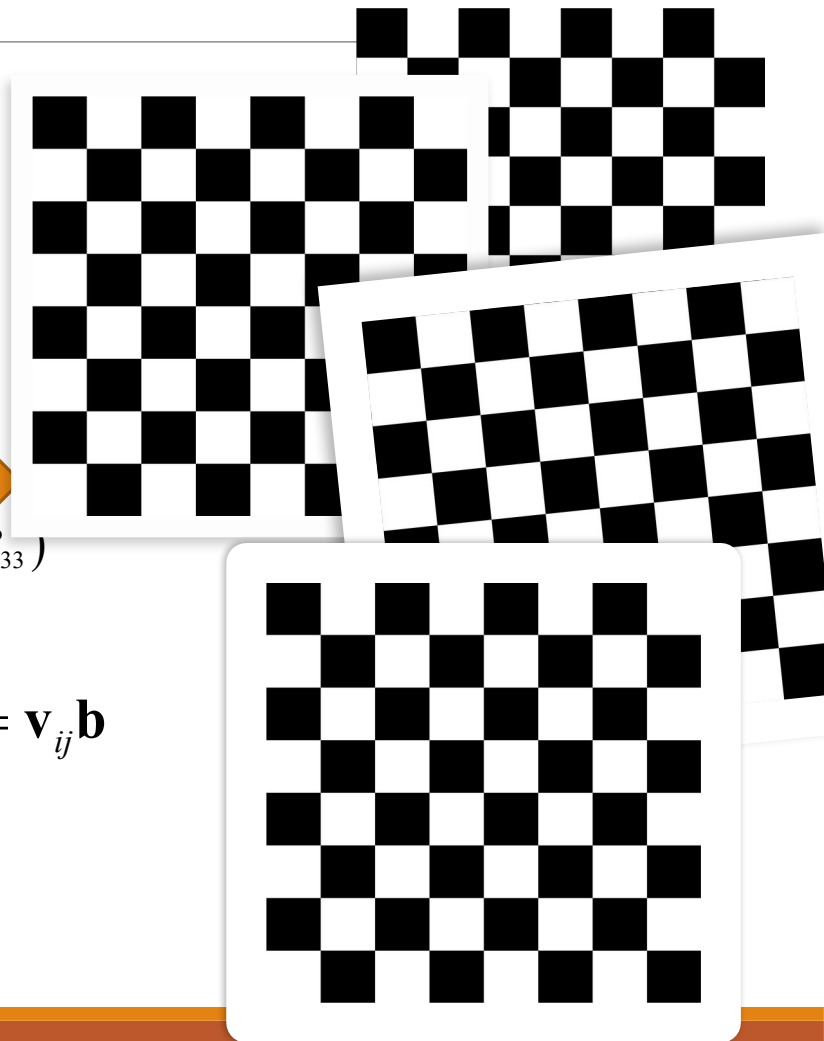
$$\mathbf{h}_1^T A^{-T} A^{-1} \mathbf{h}_2 = 0$$

$$\mathbf{h}_1^T A^{-T} A^{-1} \mathbf{h}_1 = \mathbf{h}_2^T A^{-T} A^{-1} \mathbf{h}_2 \quad \mathbf{b} = (B_{11}, B_{12}, B_{22}, B_{13}, B_{23}, B_{33})$$

$$B = A^{-T} A^{-1}$$

$$\begin{pmatrix} \mathbf{v}_{12}^T \\ (\mathbf{v}_{11} - \mathbf{v}_{22})^T \end{pmatrix} \mathbf{b} = 0$$

$$\mathbf{h}_i^T B \mathbf{h}_j = \mathbf{v}_{ij} \mathbf{b}$$



Zhang's Camera Calibration Method

After solving for $\mathbf{b}$, the intrinsic and extrinsic parameter matrices of the camera can be expressed as:

$$A_{11} = \sqrt{v/B_{11}},$$

$$A_{12} = -B_{12}A_{11}^2/A_{22}/v,$$

$$A_{13} = A_{12}A_{23}/A_{22} - B_{13}A_{11}^2/v,$$

$$A_{22} = \sqrt{vB_{11}/(B_{11}B_{22} - B_{12}^2)},$$

$$A_{23} = (B_{12}B_{13} - B_{11}B_{23})/(B_{11}B_{12} - B_{12}^2),$$

$$v = B_{33} - [B_{13}^2 + v_0(B_{12}B_{13} - B_{11}B_{23})]/B_{11}$$

$$\mathbf{r}_1 = \lambda A^{-1} \mathbf{h}_1,$$

$$\mathbf{r}_2 = \lambda A^{-1} \mathbf{h}_2,$$

$$\mathbf{r}_3 = \mathbf{r}_1 \times \mathbf{r}_2,$$

$$\mathbf{t} = \lambda A^{-1} \mathbf{h}_3.$$

Zhang's Camera Calibration Method

Bundle Adjustment

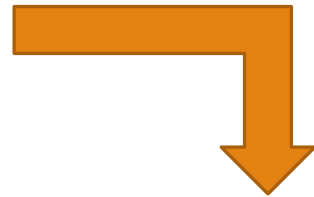
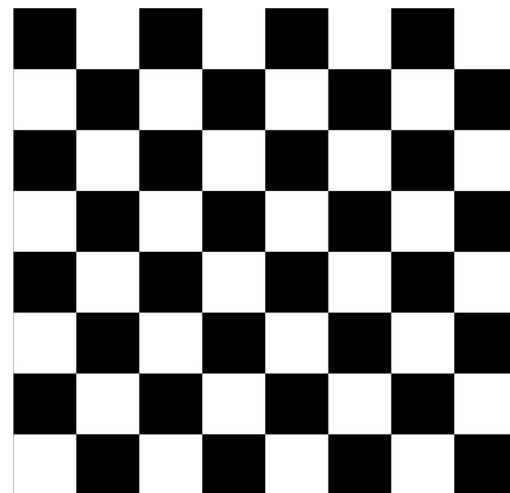
Assume there are N 3D points in the world coordinate system, denoted as ${}^W\mathbf{x}_i$, $i=1, 2, \dots, N$, corresponding to N 2D points in K images, each denoted as ${}^{S_i}\mathbf{x}_k$, $i=1,2,\dots, N$, $k=1,2,\dots,K$. Further assume the camera's intrinsic parameters are represented by $\{c_j\}_i$ and extrinsic parameters by ${}^{C_k}_wT$, $k=1,\dots,K$. Then bundle adjustment can be formulated as the following minimization problem of reprojection error:

$$E_B = \sum_{i=1}^N \sum_{k=1}^K \left\| {}^{S_i}_{I_i}T^{-1} \left(P \left({}^{C_k}_wT, \{c_j\}_i, {}^W\mathbf{x}_k \right) - {}^{S_i}\mathbf{x}_k \right) \right\|$$

Zhang's Camera Calibration Method

By fixing the world coordinate system on the calibration object, we set $Z=0$, meaning that the homogeneous coordinates of the object in the world coordinate system can be represented as $\tilde{\mathbf{M}}=(X,Y,0,1)$. The corresponding homogeneous coordinates of the pixel point can be expressed as $\tilde{\mathbf{m}}=(u,v,1)$. Therefore, according to the perspective projection model, we obtain:

$$s \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = A[R | \mathbf{t}] \begin{pmatrix} X \\ Y \\ 0 \\ 1 \end{pmatrix} = A[\mathbf{r}_1 | \mathbf{r}_2 | \mathbf{t}] \begin{pmatrix} X \\ Y \\ 1 \end{pmatrix}$$



$$\tilde{\mathbf{m}} = H\tilde{\mathbf{M}} \quad \text{with} \quad H = A[\mathbf{r}_1 \quad \mathbf{r}_2 \quad \mathbf{t}]$$

Zhang's Camera Calibration Method

$$A = \begin{bmatrix} 2725.29 & 0 & 1598.55 \\ 0 & 2640.33 & 1210.28 \\ 0 & 0 & 1 \end{bmatrix}$$

$$d_p = [-0.131475 \quad 3.66288 \quad -0.00017672 \quad 0.00136393 \quad -18.6243]$$

